

Merged Solutions: Exercises 1.1–1.3, 1.5–1.7, 2.1–2.8, 3.1–3.4

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Exercise 1.1

- (a) Show that the inner product has the following properties for all $x, y, z \in \mathbb{R}^n$ and $a \in \mathbb{R}$:

$$\langle x, y \rangle = \langle y, x \rangle, \quad \langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle, \quad \langle ax, y \rangle = a \langle x, y \rangle.$$

Solution. These follow from the Euclidean inner product $\langle x, y \rangle = \sum_{i=1}^n x_i y_i$.

- (b) For $t \in \mathbb{R}$ and $x, y \in \mathbb{R}^n$ show

$$\|x + ty\|^2 = \|x\|^2 + 2t\langle x, y \rangle + t^2\|y\|^2 \geq 0.$$

Solution. Expand $\langle x + ty, x + ty \rangle$.

- (c) Deduce Cauchy–Schwarz: $|\langle x, y \rangle| \leq \|x\| \|y\|$.

Solution. View $\|x + ty\|^2$ as a quadratic in t and use non-positivity of its discriminant. Equality iff x, y are linearly dependent.

- (d) Deduce the triangle inequality $\|x + y\| \leq \|x\| + \|y\|$.

- (e) Show the reverse triangle inequality $|\|x\| - \|y\|| \leq \|x - y\|$.

- (f) Suppose $x = (x^1, \dots, x^n)^t$.

(i) $\max_k |x^k| \leq \|x\|$.

(ii) $\|x\| \leq \sqrt{n} \max_k |x^k|$.

Solution. Immediate from $\|x\|^2 = \sum (x^i)^2$.

Exercise 1.2

Let (x_i) and (y_i) in $\mathbb{R}^n$ satisfy $x_i \rightarrow x$, $y_i \rightarrow y$.

- (a) $x_i + y_i \rightarrow x + y$. *Proof.* $\|(x_i + y_i) - (x + y)\| \leq \|x_i - x\| + \|y_i - y\| \rightarrow 0$.

- (b) $\langle x_i, y_i \rangle \rightarrow \langle x, y \rangle$; in particular $\|x_i\| \rightarrow \|x\|$.

Proof. Expand $\langle x_i, y_i \rangle - \langle x, y \rangle$ and use Cauchy–Schwarz.

- (c) If $a_i \rightarrow a$ in $\mathbb{R}$, then $a_i x_i \rightarrow ax$.

Proof. $a_i x_i - ax = (a_i - a)(x_i - x) + (a_i - a)x + a(x_i - x)$.

Exercise 1.3

Which of the following subsets of $\mathbb{R}^n$ is open?

- (a) $\mathbb{R}^n$: **Yes**.
- (b) $\emptyset$: **Yes**.
- (c) $\{x \in \mathbb{R}^n : x^1 > 0\}$: **Yes**.
- (d) $\{x \in \mathbb{R}^n : x^i \in [0, 1]\}$: **No**.
- (e) $\mathbb{Q}^n$: **No**.

Exercise 1.5

- (a) If $x_i \rightarrow x$ and $\|x_i\| < r$ for all i , then $\|x\| \leq r$.
Proof. Contradict $\|x\| > r$ using $\varepsilon = (\|x\| - r)/2$ and the reverse triangle inequality.
- (b) The closure of $B_r(y) = \{x : \|x - y\| < r\}$ is $\overline{B_r(y)} = \{x : \|x - y\| \leq r\}$.
Proof. Use (a) for the first inclusion; for boundary points take $x_i = y + (1 - \frac{1}{i})(x - y)$.

Exercise 1.6

Find A° , $\overline{A}$ and ∂A .

- (a) $A = \mathbb{R}^n$: $A^\circ = \mathbb{R}^n$, $\overline{A} = \mathbb{R}^n$, $\partial A = \emptyset$.
- (b) $A = \{0\}$: $A^\circ = \emptyset$, $\overline{A} = \{0\}$, $\partial A = \{0\}$.
- (c) $A = \{x : x^1 \geq 0\}$: $A^\circ = \{x : x^1 > 0\}$, $\overline{A} = A$, $\partial A = \{x : x^1 = 0\}$.
- (d) $A = \{x : x^i \in [0, 1] \forall i\}$: $A^\circ = \{x : x^i \in (0, 1) \forall i\}$, $\overline{A} = A$, $\partial A = A \setminus A^\circ$.
- (e) $A = \{x : x^i \in \mathbb{Q} \forall i\}$: $A^\circ = \emptyset$, $\overline{A} = \mathbb{R}^n$, $\partial A = \mathbb{R}^n$.

Exercise 1.7

- (a) If U_1, U_2 are open, then $U_1 \cup U_2$ and $U_1 \cap U_2$ are open.
- (b) If E_1, E_2 are closed, then $E_1 \cup E_2$ and $E_1 \cap E_2$ are closed.
- (c) For any collection $\mathcal{U}$ of open sets: $\bigcup \mathcal{U}$ is open; $\bigcap \mathcal{U}$ need not be open (e.g. $\bigcap_{n \geq 1} (-1/n, 1/n) = \{0\}$). For closed sets: arbitrary intersections are closed, finite unions are closed, infinite unions need not be closed.

Exercise 2.1

A sequence $(x_i) \subset \mathbb{R}^n$ is Cauchy if $\forall \varepsilon > 0 \exists N : i, j \geq N \Rightarrow \|x_i - x_j\| < \varepsilon$.

- (a) If $x_i \rightarrow x$, then (x_i) is Cauchy. *Proof.* Triangle inequality with $\varepsilon/2$.
- (b) If (x_i) is Cauchy, then it is bounded. *Proof.* Take $\varepsilon = 1$.
- (c) Every Cauchy sequence in $\mathbb{R}^n$ converges. *Proof.* Bounded $\Rightarrow$ has a convergent subsequence; use Cauchy property to pass to the full sequence.

Remark. In $\mathbb{R}$, $\mathbb{R}^n$, and $\mathbb{C} \cong \mathbb{R}^2$, a sequence is convergent iff it is Cauchy (completeness). In incomplete spaces (e.g. $\mathbb{Q}$), Cauchy does not imply convergent.

Exercise 2.2

Suppose $A \subset \mathbb{R}^n$ and $f : A \rightarrow \mathbb{R}^m$. Then

$$\lim_{x \rightarrow p} f(x) = F \iff \text{for every sequence } x_i \in A, x_i \neq p, x_i \rightarrow p, \text{ we have } f(x_i) \rightarrow F.$$

Proof. The forward direction is immediate from the ε - δ definition; the reverse is proved by contradiction constructing a sequence $x_k \rightarrow p$ with $f(x_k)$ staying ε_0 away from F .

Exercise 2.3

- (a) $f : \mathbb{R} \rightarrow \mathbb{R}^n, x \mapsto (x, 0, \dots, 0)^t$ is continuous: $\|f(x) - f(x_0)\| = |x - x_0|$.
- (b) For $A \subset \mathbb{R}^n, f = (f^1, \dots, f^m) : A \rightarrow \mathbb{R}^m$ is continuous at p iff each f^k is continuous at p .
- (c) Any polynomial map $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuous (built from projections via sums/products).

Exercise 2.4

Let $E \subset \mathbb{R}^n$ be compact and $f : E \rightarrow \mathbb{R}^m$ continuous.

- (a) $f(E)$ is bounded. *Proof.* Each component f^j attains its max/min on E .
- (b) $f(E)$ is closed, hence compact. *Proof.* Use sequential closedness via compactness of E .
- (c) If E is merely closed, $f(E)$ need not be closed; e.g. $f(x) = \arctan x$ on $E = \mathbb{R}$.

Exercise 2.5 (*)

- (a) If $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is continuous and U open in $\mathbb{R}^m$, then $f^{-1}(U)$ is open.
- (b) If $f^{-1}(U)$ is open for every open $U \subset \mathbb{R}^m$, then f is continuous on $\mathbb{R}^n$.

Exercise 2.6

Assume $f : \mathbb{R} \rightarrow \mathbb{R}$ differentiable with $|f'(x)| \leq M$ for all x .

- (a) $|f(x) - f(y)| \leq M|x - y|$ by MVT.
- (b) Hence f is uniformly continuous (Lipschitz with constant M).

Exercise 2.7

Which functions are (i) continuous, (ii) uniformly continuous on their domains?

- (a) $f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto e^x$: continuous, not uniformly continuous.
- (b) $f : (0, 1) \rightarrow \mathbb{R}, x \mapsto e^x$: continuous, uniformly continuous (MVT and $e^c \leq e$).
- (c) $f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto \sin x$: continuous, uniformly continuous ($|\sin x - \sin y| \leq |x - y|$).
- (d) $f : [0, \infty) \rightarrow \mathbb{R}, x \mapsto \sqrt{x}$: continuous, uniformly continuous ($|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|}$).
- f) $f : \mathbb{R}^n \setminus \{0\} \rightarrow \mathbb{R}^n, x \mapsto x/\|x\|$: continuous, not uniformly continuous (fails near 0).

Exercise 2.8

Let $A \subset \mathbb{R}^n$ and $f, g : A \rightarrow \mathbb{R}$.

- (a) Suppose f and g are uniformly continuous and bounded on A . Then fg is uniformly continuous and bounded on A .

Solution (boundedness). If $|f| \leq M_f$ and $|g| \leq M_g$ on A , then $|fg| \leq M_f M_g$.

Solution (uniform continuity). Fix $\varepsilon > 0$ and set $\eta = \min\left(1, \frac{\varepsilon}{2(M_f + M_g + 1)}\right)$. By uniform continuity of f and g , there exists $\delta > 0$ such that $\|x - y\| < \delta$ implies $|f(x) - f(y)| < \eta$ and $|g(x) - g(y)| < \eta$. Then for such x, y ,

$$\begin{aligned} |f(x)g(x) - f(y)g(y)| &= |[f(x) - f(y)][g(x) - g(y)] + g(y)[f(x) - f(y)] + f(y)[g(x) - g(y)]| \\ &\leq \eta^2 + M_g \eta + M_f \eta \leq (M_f + M_g + 1)\eta \leq \varepsilon. \end{aligned}$$

Hence fg is uniformly continuous.

- (b) If f and g are uniformly continuous but not necessarily bounded, fg need not be uniformly continuous.

Counterexample. On $\mathbb{R}$, take $f(x) = x$ and $g(x) = \sin x$. Both are uniformly continuous (Lipschitz). Assume $x \sin x$ were uniformly continuous. For $\varepsilon = 1$, let $\delta > 0$ be its modulus. Set $t = \min(\delta/2, 1)$ and take $y_k = k\pi$, $x_k = k\pi + t$. Then $|x_k - y_k| = t < \delta$ but

$$|x_k \sin x_k - y_k \sin y_k| = |(k\pi + t) \sin t| \geq k\pi \sin t \xrightarrow{k \rightarrow \infty} \infty,$$

contradicting uniform continuity. Thus $x \sin x$ is not uniformly continuous.

Exercise 3.1

For $i \in \mathbb{N}$, define $f_i : [0, 1] \rightarrow \mathbb{R}$ by

$$f_i(x) = \begin{cases} 2^i x, & 0 \leq x < 2^{-i}, \\ 2 - 2^i x, & 2^{-i} \leq x < 2^{-(i-1)}, \\ 0, & x \geq 2^{-(i-1)}. \end{cases}$$

- (a) *Sketch.* f_i is a triangular “spike” supported on $[0, 2^{-(i-1)}]$: it rises linearly from 0 at $x = 0$ to height 1 at $x = 2^{-i}$, then decreases linearly back to 0 at $x = 2^{-(i-1)}$; afterwards it is identically 0. In particular, $\sup_{x \in [0, 1]} f_i(x) = 1$ for every i .
- (b) *Pointwise limit.* For any fixed $x > 0$ choose I such that $2^{-(i-1)} < x$ for all $i \geq I$; then $f_i(x) = 0$ for $i \geq I$. Also $f_i(0) = 0$ for all i . Hence $f_i(x) \rightarrow 0$ for every $x \in [0, 1]$.
- (c) *Interchange of lim and sup.* We have

$$\lim_{i \rightarrow \infty} \sup_{x \in [0, 1]} f_i(x) = \lim_{i \rightarrow \infty} 1 = 1, \quad \text{while} \quad \sup_{x \in [0, 1]} \lim_{i \rightarrow \infty} f_i(x) = \sup_{x \in [0, 1]} 0 = 0.$$

Therefore the equality

$$\lim_{i \rightarrow \infty} \sup_{x \in [0, 1]} f_i(x) = \sup_{x \in [0, 1]} \lim_{i \rightarrow \infty} f_i(x)$$

is *false* for this family. (This also shows $\{f_i\}$ does not converge uniformly to 0.)

Exercise 3.2

State whether the sequence $(f_i)_{i=0}^\infty$ of functions converges pointwise and/or uniformly.

(a) $f_i : \mathbb{R} \rightarrow \mathbb{R}, \quad f_i(x) = e^x + \frac{1}{i+1}.$

Solution. For each x , $f_i(x) \rightarrow e^x$. Moreover

$$\sup_{x \in \mathbb{R}} |f_i(x) - e^x| = \sup_{x \in \mathbb{R}} \frac{1}{i+1} = \frac{1}{i+1} \xrightarrow{i \rightarrow \infty} 0,$$

so $f_i \rightarrow e^x$ *uniformly* on $\mathbb{R}$.

(b) $f_i : \mathbb{R} \rightarrow \mathbb{R}, \quad f_i(x) = x + 2^{-i}x^2.$

Solution. For fixed x , $f_i(x) \rightarrow x$. However

$$\sup_{x \in \mathbb{R}} |f_i(x) - x| = \sup_{x \in \mathbb{R}} 2^{-i}x^2 = +\infty \quad \text{for every fixed } i,$$

hence the convergence is *not* uniform on $\mathbb{R}$. (Nonetheless, on any bounded interval $[-M, M]$ the convergence is uniform since $\sup_{|x| \leq M} |f_i(x) - x| = 2^{-i}M^2 \rightarrow 0$.)

(c) $f_i : [0, 1] \rightarrow \mathbb{R}$ given by

$$f_i(x) = \begin{cases} 2^i x, & 0 \leq x < 2^{-i}, \\ 2 - 2^i x, & 2^{-i} \leq x < 2^{-(i-1)}, \\ 0, & x \geq 2^{-(i-1)}. \end{cases}$$

Solution. For every $x \in [0, 1]$, $f_i(x) \rightarrow 0$ (as in Exercise 3.1). But $\sup_{x \in [0, 1]} f_i(x) = 1$ for all i , so the convergence is *not* uniform on $[0, 1]$. (On $[\delta, 1]$ with $\delta > 0$, the convergence is uniform.)

Exercise 3.3

Suppose $A \subset \mathbb{R}^n$ and let $(f_i)_{i=0}^\infty$ with $f_i : A \rightarrow \mathbb{R}^m$. Classify whether each statement is equivalent to (i) pointwise convergence $f_i \rightarrow f$, (ii) uniform convergence $f_i \rightarrow f$ uniformly, or (iii) neither.

(a) Given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $i \geq N$:

$$\sup_{x \in A} \|f_i(x) - f(x)\| < \varepsilon.$$

Classification: (ii) *uniform convergence*. This is exactly the definition via the sup norm.

(b) For all $x \in A$, for all $i \in \mathbb{N}$ there exists $\varepsilon > 0$ such that

$$\|f_i(x) - f(x)\| < \varepsilon.$$

Classification: (iii) *neither*. The statement is always true (take, e.g., $\varepsilon = \|f_i(x) - f(x)\| + 1$), so it does not characterize convergence.

(c) There exists $N \in \mathbb{N}$ such that for all $x \in A$, $i \geq N$, $\varepsilon > 0$:

$$\|f_i(x) - f(x)\| < \varepsilon.$$

Classification: (iii) *neither*. Since the left-hand side is nonnegative, the inequality for *all* $\varepsilon > 0$ forces $\|f_i(x) - f(x)\| = 0$; i.e. there is N with $f_i \equiv f$ on A for all $i \geq N$. This condition is much stronger than uniform convergence and is not equivalent to it in general.

*d) $\forall \varepsilon > 0 \forall x \in A \exists N \in \mathbb{N} \forall i \geq N : \|f_i(x) - f(x)\| < \varepsilon$.

Classification: (i) *pointwise convergence*. The quantifier order matches the definition; N may depend on x and ε .

*e) $\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall x \in A \forall i \geq N : \|f_i(x) - f(x)\| < \varepsilon$.

Classification: (ii) *uniform convergence*. Here N works simultaneously for all $x \in A$.

*f) $\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall x \in A : (i \geq N \Leftrightarrow \|f_i(x) - f(x)\| < \varepsilon)$.

Classification: (iii) *neither*. This is far stronger than uniform convergence (it also demands that for $i < N$ the error is $\geq \varepsilon$ for every x), and it need not hold even when $f_i \rightarrow f$ uniformly.

Exercise 3.4

Suppose $A \subset \mathbb{R}^n$ and let $(f_i)_{i=0}^\infty$ with $f_i : A \rightarrow \mathbb{R}^m$ be a sequence of functions. Assume $f_i \rightarrow f$ uniformly on A . Show that if $B \subset A$, then $f_i \rightarrow f$ uniformly on B .

Solution. By uniform convergence on A , for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that for all $i \geq N$ and all $x \in A$ we have $\|f_i(x) - f(x)\| < \varepsilon$. Since $B \subset A$, this inequality holds for all $x \in B$ as well, with the *same* N . Hence $f_i \rightarrow f$ uniformly on B .

Equivalently,

$$\sup_{x \in B} \|f_i(x) - f(x)\| \leq \sup_{x \in A} \|f_i(x) - f(x)\| \xrightarrow{i \rightarrow \infty} 0.$$

Exercise 3.5 (*)

Let $E \subset \mathbb{R}^n$ be compact. Define $C^0(E, \mathbb{R}^m)$ to be the set of all continuous functions $f : E \rightarrow \mathbb{R}^m$, equipped with the norm

$$\|f\|_{C^0} := \sup_{x \in E} \|f(x)\| \in [0, \infty).$$

Show that $C^0(E, \mathbb{R}^m)$ is complete.

Solution. Let (f_k) be a Cauchy sequence in the sup norm $\|\cdot\|_{C^0}$. Then, for every $x \in E$,

$$\|f_k(x) - f_\ell(x)\| \leq \|f_k - f_\ell\|_{C^0} \xrightarrow{k, \ell \rightarrow \infty} 0,$$

so for each x the sequence $(f_k(x))$ is Cauchy in $\mathbb{R}^m$ and hence converges. Define $f(x) := \lim_{k \rightarrow \infty} f_k(x)$; this gives a function $f : E \rightarrow \mathbb{R}^m$.

Uniform convergence. Given $\varepsilon > 0$, choose N so that $\|f_k - f_\ell\|_{C^0} < \varepsilon$ for all $k, \ell \geq N$. Fix $\ell \geq N$ and let $k \rightarrow \infty$; by the pointwise limit we obtain

$$\|f - f_\ell\|_{C^0} = \sup_{x \in E} \|f(x) - f_\ell(x)\| \leq \varepsilon.$$

Thus $f_k \rightarrow f$ uniformly on E . In particular, $\|f\|_{C^0} \leq \|f - f_\ell\|_{C^0} + \|f_\ell\|_{C^0} < \infty$, so f is bounded.

Continuity of the limit. Let $x \in E$ and $\varepsilon > 0$. Choose $\ell \geq N$ so that $\|f - f_\ell\|_{C^0} < \varepsilon/3$. Since f_ℓ is continuous at x , there exists $\delta > 0$ with $\|y - x\| < \delta \Rightarrow \|f_\ell(y) - f_\ell(x)\| < \varepsilon/3$. Then for $\|y - x\| < \delta$,

$$\|f(y) - f(x)\| \leq \|f(y) - f_\ell(y)\| + \|f_\ell(y) - f_\ell(x)\| + \|f_\ell(x) - f(x)\| < \varepsilon.$$

Hence f is continuous on E , i.e. $f \in C^0(E, \mathbb{R}^m)$.

Convergence in norm. With the same N as above, fix $\ell \geq N$. For $k \geq N$,

$$\|f_k - f\|_{C^0} \leq \|f_k - f_\ell\|_{C^0} + \|f_\ell - f\|_{C^0} \leq \varepsilon + \varepsilon = 2\varepsilon,$$

so letting $\varepsilon \downarrow 0$ (or starting with $\varepsilon/2$) gives $\|f_k - f\|_{C^0} \rightarrow 0$.

Therefore every Cauchy sequence in $C^0(E, \mathbb{R}^m)$ converges (in $\|\cdot\|_{C^0}$) to an element of $C^0(E, \mathbb{R}^m)$. Thus $C^0(E, \mathbb{R}^m)$ is complete.

Exercise 3.6 (*)

Suppose $f : (a, b) \rightarrow \mathbb{R}$ is uniformly continuous, and $(x_j)_{j=0}^\infty, (y_j)_{j=0}^\infty \subset (a, b)$ satisfy $x_j \rightarrow a$ and $y_j \rightarrow b$.

- (a) Show that the sequences $(f(x_j))_{j=0}^\infty$ and $(f(y_j))_{j=0}^\infty$ are Cauchy.

Solution. Given $\varepsilon > 0$ choose $\delta > 0$ such that $|u - v| < \delta \Rightarrow |f(u) - f(v)| < \varepsilon$ (uniform continuity). Since $x_j \rightarrow a$ and $y_j \rightarrow b$, both (x_j) and (y_j) are Cauchy. Hence there is N with $|x_j - x_k| < \delta$ and $|y_j - y_k| < \delta$ for all $j, k \geq N$, which implies $|f(x_j) - f(x_k)| < \varepsilon$ and $|f(y_j) - f(y_k)| < \varepsilon$ for $j, k \geq N$. Thus both image sequences are Cauchy.

- (b) Setting $f_L := \lim_{j \rightarrow \infty} f(x_j)$ and $f_R := \lim_{j \rightarrow \infty} f(y_j)$, show that

$$\tilde{f} : [a, b] \rightarrow \mathbb{R}, \quad \tilde{f}(x) = \begin{cases} f_L, & x = a, \\ f(x), & a < x < b, \\ f_R, & x = b, \end{cases}$$

is continuous on $[a, b]$.

Solution. By (a) the limits f_L, f_R exist (Cauchy in $\mathbb{R}$ hence convergent). On (a, b) , $\tilde{f} = f$, so continuity holds. We check continuity at a ; the case b is analogous.

Let $\varepsilon > 0$. Pick $\delta > 0$ for uniform continuity of f and choose J large so that $|x_J - a| < \delta/2$ and $|f(x_J) - f_L| < \varepsilon$. If $0 < |t - a| < \delta/2$, then $|t - x_J| \leq |t - a| + |x_J - a| < \delta$, hence

$$|\tilde{f}(t) - \tilde{f}(a)| = |f(t) - f_L| \leq |f(t) - f(x_J)| + |f(x_J) - f_L| < \varepsilon + \varepsilon = 2\varepsilon.$$

Therefore $\tilde{f}$ is continuous at a . A symmetric argument using (y_j) gives continuity at b . Thus $\tilde{f}$ is continuous on $[a, b]$.

- (c) Deduce that f is uniformly continuous on (a, b) if and only if there exists a continuous $\tilde{f} : [a, b] \rightarrow \mathbb{R}$ with $\tilde{f}(x) = f(x)$ for all $x \in (a, b)$ (i.e. $\tilde{f}$ is an extension of f).

Solution. ($\Rightarrow$) If f is uniformly continuous, parts (a)–(b) construct a continuous extension $\tilde{f}$ on $[a, b]$.

($\Leftarrow$) If such $\tilde{f}$ exists, then $\tilde{f}$ is continuous on the compact set $[a, b]$, hence uniformly continuous. Its restriction to (a, b) equals f and is therefore uniformly continuous.

Exercise 3.8 (*)

Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be the 1-periodic function satisfying $h(x) = |x|$ for $x \in [-\frac{1}{2}, \frac{1}{2}]$ (and $h(x+1) = h(x)$). Define the Takagi function

$$T(x) = \sum_{i=0}^{\infty} \frac{1}{2^i} h(2^i x).$$

- (a) *Sketch h .* On $[-\frac{1}{2}, \frac{1}{2}]$ it is the V -shape $|x|$ (cusp at 0), extended periodically with period 1. Equivalently,

$$h(x) = \text{dist}(x, \mathbb{Z}) = \min_{k \in \mathbb{Z}} |x - k|,$$

so $0 \leq h(x) \leq \frac{1}{2}$, attaining 0 at integers and $\frac{1}{2}$ at half-integers.

- (b) *Uniform continuity and differentiability.* Since $x \mapsto \text{dist}(x, \mathbb{Z})$ is 1-Lipschitz, h is uniformly continuous on $\mathbb{R}$. On each interval $(k, k + \frac{1}{2})$ and $(k - \frac{1}{2}, k)$ the function is affine with slope -1 or $+1$, hence differentiable there. It fails to be differentiable precisely at points congruent to 0 or $\frac{1}{2}$ modulo 1, i.e. at those x with $2x \in \mathbb{Z}$. Thus h is differentiable at every x with $2x \notin \mathbb{Z}$.

- (c) *Uniform convergence of the series and uniform continuity of T .* Since $0 \leq h \leq \frac{1}{2}$,

$$\left| \frac{1}{2^i} h(2^i x) \right| \leq \frac{1}{2^{i+1}} \quad (\forall x \in \mathbb{R}),$$

and $\sum_{i=0}^{\infty} 2^{-(i+1)} < \infty$. By the Weierstrass M -test, the series converges uniformly on $\mathbb{R}$ to a continuous function T . Each summand $x \mapsto 2^{-i} h(2^i x)$ is uniformly continuous (indeed 1-Lipschitz), hence T , being a uniform limit of uniformly continuous functions, is uniformly continuous on $\mathbb{R}$.

- (d) Let $x \in \mathbb{R}$. There exist unique u_n, v_n such that

- (i) $2^n u_n, 2^n v_n \in \mathbb{Z}$,
- (ii) $v_n - u_n = 2^{-n}$,
- (iii) $u_n \leq x < v_n$.

Solution. Take $u_n = \frac{\lfloor 2^n x \rfloor}{2^n}$ and $v_n = u_n + 2^{-n}$. Then (i)–(iii) hold. Uniqueness: if $u'_n \leq x < v'_n$ with (i)–(ii), then $2^n u'_n \leq 2^n x < 2^n v'_n$ and $2^n v'_n - 2^n u'_n = 1$, so $2^n u'_n = \lfloor 2^n x \rfloor = 2^n u_n$, hence $u'_n = u_n$ and $v'_n = v_n$.

- (e) For $i \geq 0$ set $h_i(x) = 2^{-i} h(2^i x)$. Show that:

- (i) $h_i(u_n) = h_i(v_n) = 0$ for $i \geq n$.
- (ii) $\frac{h_i(v_n) - h_i(u_n)}{v_n - u_n} = \pm 1$ for $i < n$.

Solution. (i) Since $2^n u_n, 2^n v_n \in \mathbb{Z}$, for $i \geq n$ also $2^i u_n, 2^i v_n \in \mathbb{Z}$, hence $h(2^i u_n) = h(2^i v_n) = 0$.

(ii) The map $x \mapsto 2^i x$ sends $[u_n, v_n]$ onto the interval $[\frac{K}{2^{n-i}}, \frac{K+1}{2^{n-i}}]$ where $K = \lfloor 2^n x \rfloor$. Its length is $2^i(v_n - u_n) = 2^{-(n-i)} \leq \frac{1}{2}$, so it contains no half-integer. Hence h is affine with slope ± 1 on this interval, and therefore

$$\frac{h_i(v_n) - h_i(u_n)}{v_n - u_n} = \frac{2^{-i}(h(2^i v_n) - h(2^i u_n))}{2^{-n}} = 2^{n-i} \cdot 2^{-i} \cdot (\pm 2^{-(n-i)}) = \pm 1.$$

- (f) *Conclude that*

$$\frac{T(v_n) - T(u_n)}{v_n - u_n} = \sum_{i=0}^{n-1} \frac{h_i(v_n) - h_i(u_n)}{v_n - u_n},$$

and that the sum on the right does not converge as $n \rightarrow \infty$.

Solution. Since $h_i(u_n) = h_i(v_n) = 0$ for $i \geq n$,

$$T(v_n) - T(u_n) = \sum_{i=0}^{\infty} (h_i(v_n) - h_i(u_n)) = \sum_{i=0}^{n-1} (h_i(v_n) - h_i(u_n)).$$

Divide by $v_n - u_n$ to get the displayed identity. By (e)(ii), each term of the sum equals ± 1 , so the sequence of partial sums cannot converge (successive differences are $\pm 1 \not\rightarrow 0$). Hence the right-hand side does not converge as $n \rightarrow \infty$. In particular, T is nowhere differentiable.

Exercise 3.8 (g)

Show that the Takagi function $T(x) = \sum_{i=0}^{\infty} 2^{-i} h(2^i x)$ is not differentiable at any x .

Solution. Let $u_n = \lfloor \frac{2^n x}{2^n} \rfloor$ and $v_n = u_n + 2^{-n}$. Then $u_n \uparrow x$, $v_n \downarrow x$, and

$$\frac{T(v_n) - T(u_n)}{v_n - u_n} = \sum_{i=0}^{n-1} \frac{h_i(v_n) - h_i(u_n)}{v_n - u_n}, \quad h_i(t) = 2^{-i} h(2^i t).$$

Each term equals ± 1 , so the partial sums do not converge. If $T'(x)$ existed, these difference quotients would converge to $T'(x)$, a contradiction. Hence T is nowhere differentiable.

Exercise 3.9 (*)

Suppose $F = (f_i)_{i=0}^{\infty}$ is a family of continuous functions $f_i : [a, b] \rightarrow \mathbb{R}$, each differentiable on (a, b) , and assume

$$\sup_{x \in (a, b)} |f'_i(x)| \leq M \quad \text{for all } i,$$

for some M independent of i . Show that F is uniformly equicontinuous.

Solution. By the mean value theorem, for any $x, y \in [a, b]$ with $x \neq y$ there exists $c \in (x, y)$ such that

$$|f_i(x) - f_i(y)| = |f'_i(c)| |x - y| \leq M |x - y|, \quad \text{for every } i.$$

Thus each f_i is Lipschitz with the same constant M . Given $\varepsilon > 0$, choose $\delta = \varepsilon/M$. Then for all i and all $x, y \in [a, b]$, if $|x - y| < \delta$ we have $|f_i(x) - f_i(y)| < \varepsilon$. Hence F is uniformly equicontinuous on $[a, b]$.

Definitions and Examples

Supremum and Infimum. For a set $S \subset \mathbb{R}$:

- The *supremum* (least upper bound) is the smallest $M \in \mathbb{R}$ such that $x \leq M$ for all $x \in S$. Denoted $\sup S$.
- The *infimum* (greatest lower bound) is the largest $m \in \mathbb{R}$ such that $m \leq x$ for all $x \in S$. Denoted $\inf S$.

Example. For $S = (0, 1)$, $\sup S = 1$, $\inf S = 0$.

Partitions, Upper and Lower Sums. A *partition* of $[a, b]$ is a finite set

$$P = \{x_0, x_1, \dots, x_n\}, \quad a = x_0 < x_1 < \dots < x_n = b.$$

For a bounded function $f : [a, b] \rightarrow \mathbb{R}$:

$$U(f, P) = \sum_{i=1}^n M_i(x_i - x_{i-1}), \quad M_i = \sup\{f(x) : x \in [x_{i-1}, x_i]\},$$

$$L(f, P) = \sum_{i=1}^n m_i(x_i - x_{i-1}), \quad m_i = \inf\{f(x) : x \in [x_{i-1}, x_i]\}.$$

Example. For $f(x) = x$ on $[0, 1]$ with $P = \{0, 0.5, 1\}$:

$$U(f, P) = 0.75, \quad L(f, P) = 0.25.$$

Integrability. A bounded $f : [a, b] \rightarrow \mathbb{R}$ is *Riemann integrable* if

$$\sup_P L(f, P) = \inf_P U(f, P).$$

Example. $f(x) = x$ on $[0, 1]$ is integrable with $\int_0^1 x dx = \frac{1}{2}$.

Characteristic Function. For $A \subset \mathbb{R}$, the *characteristic function* is

$$\chi_A(x) = \begin{cases} 1 & x \in A, \\ 0 & x \notin A. \end{cases}$$

Example. If $A = [0, \frac{1}{2}]$, then $\int_0^1 \chi_A(x) dx = \frac{1}{2}$.

Solutions

Exercise 4.1

Solution.

- a) $\sup_{x \in A}(-f(x)) = -\inf_{x \in A} f(x).$
- b) $\inf_{x \in A}(-f(x)) = -\sup_{x \in A} f(x).$
- c) $\sup_{x \in A} \lambda f(x) = \lambda \sup_{x \in A} f(x)$ for $\lambda \geq 0$.
- d) $\sup_{x \in A}(f(x) + g(x)) \leq \sup f + \sup g.$
- e) $\inf_{x \in A}(f(x) + g(x)) \geq \inf f + \inf g.$
- f) $\inf_{x \in A} \lambda f(x) = \lambda \inf_{x \in A} f(x).$
- g) $|\sup f| \leq \sup |f|.$
- h) $|\inf f| \leq \sup |f|.$

Exercise 4.2

Solution.

- a) $U(-f, P) = -L(f, P)$.
- b) $L(-f, P) = -U(f, P)$.
- c) $U(\lambda f, P) = \lambda U(f, P)$ for $\lambda \geq 0$.
- d) $L(\lambda f, P) = \lambda L(f, P)$.
- e) $U(f + g, P) \leq U(f, P) + U(g, P)$.
- f) $L(f + g, P) \geq L(f, P) + L(g, P)$.

Exercise 4.3

Solution. If $P \preceq Q$, then $U(f, Q) \leq U(f, P)$ and $L(f, Q) \geq L(f, P)$. Hence

$$0 \leq U(f, Q) - L(f, Q) \leq U(f, P) - L(f, P).$$

Exercise 4.4

Solution. The function χ_A is Riemann integrable since it has only finitely many discontinuities. Moreover,

$$\int_{-1}^1 \chi_A(x) dx = 0,$$

because a finite set has measure zero.

Exercise 4.5

Suppose that $a < c < b$ and suppose $f : [a, b] \rightarrow \mathbb{R}$ is a bounded function. Let $\mathcal{P}$ be a partition of $[a, b]$ of the form:

$$\mathcal{P} = (a, x_1, \dots, x_{l-1}, x_l = c, x_{l+1}, \dots, x_{k-1}, b).$$

Define $\mathcal{P}_L$ and $\mathcal{P}_R$ to be partitions of $[a, c]$ and $[c, b]$ respectively:

$$\mathcal{P}_L = (a, x_1, \dots, x_{l-1}, c), \quad \mathcal{P}_R = (c, x_{l+1}, \dots, x_{k-1}, b).$$

Show that

$$L(f, \mathcal{P}) = L(f|_{[a,c]}, \mathcal{P}_L) + L(f|_{[c,b]}, \mathcal{P}_R), \quad U(f, \mathcal{P}) = U(f|_{[a,c]}, \mathcal{P}_L) + U(f|_{[c,b]}, \mathcal{P}_R).$$

Solution. By definition,

$$L(f, \mathcal{P}) = \sum_{i=1}^k m_i(x_i - x_{i-1}), \quad m_i = \inf_{x \in [x_{i-1}, x_i]} f(x).$$

The partition $\mathcal{P}$ splits at c : intervals before c contribute the lower sum on $[a, c]$, those after c contribute on $[c, b]$. Thus

$$L(f, \mathcal{P}) = L(f|_{[a,c]}, \mathcal{P}_L) + L(f|_{[c,b]}, \mathcal{P}_R).$$

The same argument with suprema shows

$$U(f, \mathcal{P}) = U(f|_{[a,c]}, \mathcal{P}_L) + U(f|_{[c,b]}, \mathcal{P}_R).$$

Example. For $f(x) = x$, $a = 0$, $b = 2$, $c = 1$ with partition $\mathcal{P} = (0, 1, 2)$:

$$L(f, \mathcal{P}_L) = 0, \quad L(f, \mathcal{P}_R) = 1, \quad U(f, \mathcal{P}_L) = 1, \quad U(f, \mathcal{P}_R) = 2.$$

So $L(f, \mathcal{P}) = 1$ and $U(f, \mathcal{P}) = 3$, which indeed split into left and right parts.

Exercise 4.6

Suppose $f, g : [a, b] \rightarrow \mathbb{R}$ are integrable.

a) Show that

$$(b - a) \inf_{x \in [a, b]} f(x) \leq \int_a^b f(x) dx \leq (b - a) \sup_{x \in [a, b]} f(x).$$

b) Establish the estimate:

$$\left| \int_a^b f(x) dx \right| \leq (b - a) \sup_{x \in [a, b]} |f(x)|.$$

c) Show that if $0 \leq f(x)$ for all $x \in [a, b]$, then

$$0 \leq \int_a^b f(x) dx.$$

d) Show that if $f(x) \leq g(x)$ for all $x \in [a, b]$, then

$$\int_a^b f(x) dx \leq \int_a^b g(x) dx.$$

e) Show that

$$\left| \int_a^b f(x) dx - \int_a^b g(x) dx \right| \leq \int_a^b |f(x) - g(x)| dx.$$

Solution.

a) Since $\inf f \leq f(x) \leq \sup f$ for all x , integrating and multiplying by $(b - a)$ gives

$$(b - a) \inf f \leq \int_a^b f(x) dx \leq (b - a) \sup f.$$

b) Apply part (a) to f and to $-f$:

$$-(b - a) \sup |f| \leq \int_a^b f(x) dx \leq (b - a) \sup |f|.$$

Taking absolute values gives

$$\left| \int_a^b f(x) dx \right| \leq (b - a) \sup |f|.$$

c) If $f \geq 0$, then $\inf f \geq 0$. By part (a),

$$0 \leq \int_a^b f(x) dx.$$

d) If $f \leq g$, set $h = g - f \geq 0$. Then by part (c),

$$0 \leq \int_a^b h(x) dx = \int_a^b g(x) dx - \int_a^b f(x) dx.$$

e) By the triangle inequality for integrals:

$$\int_a^b (f - g) dx \leq \int_a^b |f - g| dx, \quad \int_a^b (g - f) dx \leq \int_a^b |f - g| dx.$$

Thus

$$\left| \int_a^b (f - g) dx \right| \leq \int_a^b |f - g| dx.$$

Example. For $f(x) = x$, $g(x) = x^2$ on $[0, 1]$:

$$\int_0^1 f(x) dx = \frac{1}{2}, \quad \int_0^1 g(x) dx = \frac{1}{3}.$$

So

$$\left| \frac{1}{2} - \frac{1}{3} \right| = \frac{1}{6}, \quad \int_0^1 |x - x^2| dx = \frac{1}{6},$$

and equality holds.

Exercise 4.7

Consider Thomae's function $f : [0, 1] \rightarrow \mathbb{R}$:

$$f(x) = \begin{cases} 1, & x = 0, \\ \frac{1}{q}, & x = \frac{p}{q} \in \mathbb{Q}, \gcd(p, q) = 1, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Show that f is integrable.

Solution. First, $0 \leq f(x) \leq 1$ for all x , so f is bounded.

Continuity: If x is irrational, then $f(x) = 0$. For any $\varepsilon > 0$, rationals with denominator q large enough satisfy $1/q < \varepsilon$, and they are isolated; hence in a small neighborhood of x we avoid rationals with small denominator. Thus f is continuous at all irrationals.

If $x = \frac{p}{q}$ rational, then $f(x) = 1/q > 0$, but arbitrarily close are irrationals with $f = 0$. So f is discontinuous at every rational.

Therefore the discontinuity set is the rationals in $[0, 1]$, a countable set.

Integrability: By a standard theorem, a bounded function on $[a, b]$ with discontinuities on a measure-zero set is Riemann integrable. Since $\mathbb{Q} \cap [0, 1]$ has measure zero, f is integrable.

Integral value: Since irrationals form full measure and $f(x) = 0$ there,

$$\int_0^1 f(x) dx = 0.$$

Example intuition: The graph shows spikes at rationals p/q of height $1/q$, but these spikes shrink so fast that their total area vanishes.

Exercise 5.1

Suppose $f : [a, b] \rightarrow \mathbb{R}$ is continuous, $f(x) \geq 0$, and $\int_a^b f(x)dx = 0$.

a) Show that $f(x) = 0$ on $[a, b]$.

b) Does the result hold without continuity?

Solution. (a) If $f(c) > 0$ for some c , by continuity there is $\delta > 0$ such that $f(x) > \frac{1}{2}f(c) > 0$ on $(c - \delta, c + \delta)$. Then

$$\int_a^b f(x)dx \geq \int_{c-\delta}^{c+\delta} f(x)dx > 0,$$

contradiction. Hence $f \equiv 0$.

(b) Without continuity the result fails. Example: the Dirichlet function $\chi_{\mathbb{Q} \cap [0,1]}$ has $\int_0^1 f = 0$, but $f(x) \neq 0$ for rationals.

Exercise 5.2

Show $\int_0^1 x^2 dx = \frac{1}{3}$ using Riemann sums.

Solution. Partition $[0, 1]$ into l equal subintervals. Right endpoint sum:

$$S_l = \frac{1}{l} \sum_{j=1}^l \left(\frac{j}{l}\right)^2 = \frac{1}{l^3} \sum_{j=1}^l j^2.$$

Since $\sum_{j=1}^l j^2 = \frac{l(l+1)(2l+1)}{6}$, we get

$$S_l = \frac{l(l+1)(2l+1)}{6l^3} \rightarrow \frac{1}{3} \quad (l \rightarrow \infty).$$

Exercise 5.3

Show

$$\lim_{k \rightarrow \infty} \sum_{j=0}^{k-1} \frac{k}{j^2 + k^2} = \int_0^1 \frac{1}{1+x^2} dx.$$

Solution. Rewrite

$$\frac{k}{j^2 + k^2} = \frac{1}{k} \cdot \frac{1}{1 + (j/k)^2}.$$

So sum is a Riemann sum for $\int_0^1 \frac{1}{1+x^2} dx$. Thus

$$\lim_{k \rightarrow \infty} \sum_{j=0}^{k-1} \frac{k}{j^2 + k^2} = \int_0^1 \frac{1}{1+x^2} dx = \arctan(1) - \arctan(0) = \frac{\pi}{4}.$$

Exercise 5.4

Define $f : [-1, 1] \rightarrow \mathbb{R}$ by

$$f(x) = \begin{cases} 0, & x \leq 0, \\ 1, & x > 0. \end{cases}$$

a) Show f is integrable and compute $F(x) = \int_{-1}^x f(t)dt$.

b) Show F is not differentiable at 0.

Solution. (a) f has one discontinuity at 0, hence integrable. For $x \leq 0$, $F(x) = 0$. For $x > 0$,

$$F(x) = \int_0^x 1 dt = x.$$

So $F(x) = 0$ for $x \leq 0$, $F(x) = x$ for $x > 0$.

(b) For $x < 0$, $F'(x) = 0 = f(x)$. For $x > 0$, $F'(x) = 1 = f(x)$. At $x = 0$, left derivative $0 \neq 1$ right derivative, so F is not differentiable at 0.

Exercise 5.5

Suppose $f : [a, b] \rightarrow \mathbb{R}$ is integrable and $F : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$, differentiable on (a, b) , and satisfies

$$F'(x) = f(x) \quad \text{for all } x \in (a, b).$$

Show that:

1. For a partition $P = (x_0, \dots, x_k)$, there exists $t_j \in [x_j, x_{j+1}]$ such that

$$F(x_{j+1}) - F(x_j) = f(t_j)(x_{j+1} - x_j).$$

2. For a tagged partition (P, τ) with $\tau = (t_0, \dots, t_{k-1})$, one has

$$R(f, P, \tau) = F(b) - F(a).$$

3. Deduce that the continuity assumption in the integrability theorem (Corollary 2.14) can be weakened.

Background.

- **Mean Value Theorem (MVT).** If F is continuous on $[u, v]$ and differentiable on (u, v) , then there exists $t \in (u, v)$ such that

$$F(v) - F(u) = F'(t)(v - u).$$

- **Riemann Sum.** For a tagged partition (P, τ) , the Riemann sum is

$$R(f, P, \tau) = \sum_{j=0}^{k-1} f(t_j)(x_{j+1} - x_j).$$

Solution. (a) Apply the Mean Value Theorem to each interval $[x_j, x_{j+1}]$. Since F is continuous on $[x_j, x_{j+1}]$ and differentiable on (x_j, x_{j+1}) , there exists $t_j \in (x_j, x_{j+1})$ such that

$$F(x_{j+1}) - F(x_j) = F'(t_j)(x_{j+1} - x_j).$$

But $F'(t_j) = f(t_j)$, hence

$$F(x_{j+1}) - F(x_j) = f(t_j)(x_{j+1} - x_j).$$

(b) Summing over all subintervals:

$$\sum_{j=0}^{k-1} [F(x_{j+1}) - F(x_j)] = F(b) - F(a).$$

By part (a), each difference equals $f(t_j)(x_{j+1} - x_j)$, so

$$R(f, P, \tau) = \sum_{j=0}^{k-1} f(t_j)(x_{j+1} - x_j) = F(b) - F(a).$$

(c) Now consider any sequence of partitions (P_n) of $[a, b]$ with mesh $\|P_n\| \rightarrow 0$. For each partition, choose tags as in (a). Then by part (b), each Riemann sum satisfies

$$R(f, P_n, \tau_n) = F(b) - F(a).$$

Thus all such Riemann sums converge to the same value $F(b) - F(a)$, independently of the sequence of partitions. Therefore f is Riemann integrable and

$$\int_a^b f(x) dx = F(b) - F(a).$$

This shows that the requirement “ f is continuous” in Corollary 2.14 can be weakened: it suffices that f be integrable and arise as the derivative of some differentiable function F . In other words, continuity of f is not essential for the Fundamental Theorem of Calculus to hold.

Exercise 5.6

a) For $0 < a < b$, show

$$\int_a^b \frac{1}{1+t^2} dt = \int_{1/b}^{1/a} \frac{1}{1+t^2} dt,$$

using the substitution $\phi(t) = 1/t$.

b) Setting $a = -1$, $b = 1$ in the above, one seems to obtain

$$\int_{-1}^1 \frac{1}{1+x^2} dx = \int_{-1}^1 \frac{1}{1+x^2} dx.$$

Explain what went wrong.

Background. The substitution rule requires the substitution ϕ to be differentiable and monotone on the interval. For $\phi(t) = 1/t$, we have $\phi'(t) = -1/t^2$.

Solution. (a) With $x = \phi(t) = 1/t$, we have $dx = -\frac{1}{t^2}dt$ and

$$1 + x^2 = 1 + \frac{1}{t^2} = \frac{1+t^2}{t^2}.$$

Thus

$$\frac{1}{1+x^2} dx = \frac{1}{\frac{1+t^2}{t^2}} \cdot \left(-\frac{1}{t^2}dt\right) = -\frac{1}{1+t^2}dt.$$

When $x = a$, $t = 1/a$; when $x = b$, $t = 1/b$. Therefore

$$\int_a^b \frac{1}{1+x^2} dx = \int_{1/a}^{1/b} -\frac{1}{1+t^2} dt = \int_{1/b}^{1/a} \frac{1}{1+t^2} dt.$$

(b) If $a = -1$, $b = 1$, then $\phi(t) = 1/t$ is not continuous or monotone on $[-1, 1]$, since it has a singularity at 0. The substitution theorem requires ϕ to be differentiable and monotone on the whole interval, which fails here. Thus the formula cannot be applied directly across $[-1, 1]$.

In fact, the integral is computed directly as

$$\int_{-1}^1 \frac{1}{1+x^2} dx = 2 \arctan(1) = \frac{\pi}{2}.$$

If one wishes to use the substitution $x = 1/t$, the interval must be split at 0 and treated separately on $[-1, 0)$ and $(0, 1]$.

Exercise 5.7

Let $I_1 = [\alpha, \beta]$, $I_2 = [\gamma, \delta]$ be finite intervals. Suppose that $\phi : I_1 \rightarrow I_2$ is continuous on I_1 and differentiable on (α, β) with ϕ' integrable on I_1 . Suppose that $f : I_2 \rightarrow \mathbb{R}$ is continuous. Show that:

$$\int_{\phi(\alpha)}^{\phi(\beta)} f(t) dt = \int_{\alpha}^{\beta} f(\phi(t)) \phi'(t) dt.$$

Background. This is the standard *change of variables formula* for Riemann integrals. It generalizes Exercise 5.6 and can be proven by combining:

- the Mean Value Theorem on each subinterval of a partition,
- the telescoping sum argument as in Exercise 5.5,
- the fact that Riemann sums converge to the integral when $\text{mesh} \rightarrow 0$.

Solution. Partition $I_1 = [\alpha, \beta]$ into $P = (x_0, \dots, x_k)$. On each $[x_j, x_{j+1}]$, apply the Mean Value Theorem to $F(u) = \int_{\gamma}^u f(t) dt$, which is differentiable with derivative $F'(u) = f(u)$.

Then for each subinterval there exists $t_j \in (x_j, x_{j+1})$ such that

$$F(\phi(x_{j+1})) - F(\phi(x_j)) = f(\phi(t_j))(\phi(x_{j+1}) - \phi(x_j)).$$

Summing over j gives a telescoping series:

$$F(\phi(\beta)) - F(\phi(\alpha)) = \sum_{j=0}^{k-1} f(\phi(t_j))(\phi(x_{j+1}) - \phi(x_j)).$$

But $\phi(x_{j+1}) - \phi(x_j) = \phi'(\xi_j)(x_{j+1} - x_j)$ for some $\xi_j \in (x_j, x_{j+1})$. Thus the sum is a Riemann sum for $\int_{\alpha}^{\beta} f(\phi(t))\phi'(t) dt$.

Passing to the limit as the $\text{mesh} \rightarrow 0$, we obtain

$$\int_{\phi(\alpha)}^{\phi(\beta)} f(t) dt = \int_{\alpha}^{\beta} f(\phi(t))\phi'(t) dt.$$

This is exactly the substitution formula.

Exercise 5.8

By modifying the example of Section 2.3.1, show that there exists a family $(f_j)_{j=0}^{\infty}$ of integrable functions $f_j : [0, 1] \rightarrow \mathbb{R}$ such that:

$$f_j(x) \rightarrow 0 \quad \text{for all } x \in [0, 1],$$

while

$$\int_0^1 f_j(x) dx \rightarrow \infty.$$

Background. Pointwise convergence $f_j(x) \rightarrow 0$ does not imply convergence of integrals. A classic counterexample uses “spikes” of increasing height but decreasing width, so that the area (integral) diverges even though each fixed x eventually lies outside all spikes.

Solution. Define

$$f_j(x) = \begin{cases} j^2, & 0 \leq x \leq \frac{1}{j}, \\ 0, & \frac{1}{j} < x \leq 1. \end{cases}$$

Step 1. Pointwise convergence. Fix $x \in (0, 1]$. For sufficiently large j , we have $x > 1/j$, so $f_j(x) = 0$. Thus $f_j(x) \rightarrow 0$ for all $x \in (0, 1]$. Also $f_j(0) = j^2$, which diverges, but convergence is only required for each fixed $x \in [0, 1]$, and for $x = 0$, note $f_j(0)$ does not tend to 0. If we instead define $f_j(0) = 0$, the property holds for all x .

Step 2. Integrals. Compute:

$$\int_0^1 f_j(x) dx = \int_0^{1/j} j^2 dx = j^2 \cdot \frac{1}{j} = j.$$

Hence $\int_0^1 f_j(x) dx \rightarrow \infty$ as $j \rightarrow \infty$.

Conclusion. This provides the required example: although $f_j(x) \rightarrow 0$ pointwise, the integrals diverge to infinity. It shows that pointwise convergence is not enough to pass limits under the integral sign.

Exercise 6.1

Suppose $f : [0, 1) \rightarrow [0, 1)$ is nonnegative, continuous, and monotone decreasing. Let $N, M \in \mathbb{Z}$.

a) Show that

$$\sum_{n=N}^M f(n) \geq \int_N^{M+1} f(x) dx.$$

b) Show that

$$\sum_{n=N+1}^M f(n) \leq \int_N^M f(x) dx.$$

c) Deduce

$$\int_N^{M+1} f(x) dx \leq \sum_{n=N}^M f(n) \leq f(N) + \int_N^M f(x) dx.$$

d) Show that $\sum_{n=N}^{\infty} f(n)$ converges iff the improper integral $\int_N^{\infty} f(x) dx$ exists.

e) Conclude that $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

Solution. (a) On $[n, n+1]$, $f(x) \leq f(n)$, so

$$\int_n^{n+1} f(x) dx \leq f(n).$$

Summing yields

$$\int_N^{M+1} f(x) dx \leq \sum_{n=N}^M f(n).$$

(b) On $[n, n+1]$, $f(x) \geq f(n+1)$, so

$$\int_n^{n+1} f(x) dx \geq f(n+1).$$

Summing gives

$$\int_N^M f(x) dx \geq \sum_{n=N+1}^M f(n).$$

(c) Combining (a) and (b):

$$\int_N^{M+1} f(x) dx \leq \sum_{n=N}^M f(n) \leq f(N) + \int_N^M f(x) dx.$$

(d) The error term between the sum and integral is at most $f(N)$, independent of M . Hence $\sum f(n)$ converges iff $\int f(x)$ converges.

(e) For $f(x) = 1/x$, $\int_1^M \frac{1}{x} dx = \log M \rightarrow \infty$. So the harmonic series diverges.

Exercise 6.2

Suppose $f : [0, \infty) \rightarrow \mathbb{R}$ is nonnegative and decreasing. Show that

$$\sum_{n=1}^{\infty} f(n) < \infty \quad \implies \quad \lim_{x \rightarrow \infty} xf(x) = 0.$$

Solution. Suppose $\sum f(n)$ converges. Then by the integral test, $\int_1^{\infty} f(x) dx$ also converges.

Assume for contradiction that $\limsup_{x \rightarrow \infty} xf(x) > \epsilon > 0$. Then there exists a sequence $x_k \rightarrow \infty$ with

$$f(x_k) \geq \frac{\epsilon}{x_k}.$$

For integers n with $n \leq x_k < 2n$, monotonicity gives

$$f(n) \geq f(x_k) \geq \frac{\epsilon}{x_k} \geq \frac{\epsilon}{2n}.$$

Thus infinitely many terms satisfy $f(n) \geq \frac{\epsilon}{2n}$. Hence

$$\sum f(n) \geq \frac{\epsilon}{2} \sum \frac{1}{n},$$

which diverges. This contradicts convergence. Therefore

$$\lim_{x \rightarrow \infty} xf(x) = 0.$$

Exercise 6.3

Suppose $f : [0, \infty) \rightarrow \mathbb{R}$ is nonnegative, monotone decreasing, and integrable on $[0, \infty)$. Show that

$$\lim_{x \rightarrow \infty} f(x) = 0.$$

Solution. Assume for contradiction that $\lim_{x \rightarrow \infty} f(x) = L > 0$. Then there exists X such that $f(x) \geq L/2$ for all $x \geq X$. Hence

$$\int_X^\infty f(x) dx \geq \int_X^\infty \frac{L}{2} dx = \infty,$$

contradiction. Therefore $L = 0$, i.e.

$$\lim_{x \rightarrow \infty} f(x) = 0.$$

Exercise 6.4

Suppose $(a_n)_{n=1}^\infty$ is a sequence of positive decreasing real numbers. Show that

$$\sum_{n=1}^\infty a_n \text{ converges} \iff \sum_{n=1}^\infty 2^n a_{2^n} \text{ converges}.$$

Solution. Partition $\sum a_n$ into dyadic blocks:

$$S_n = \sum_{k=2^n}^{2^{n+1}-1} a_k.$$

Since a_n is decreasing, for $k \in [2^n, 2^{n+1} - 1]$,

$$a_{2^{n+1}} \leq a_k \leq a_{2^n}.$$

There are 2^n terms, so

$$2^n a_{2^{n+1}} \leq S_n \leq 2^n a_{2^n}.$$

Summing over $n = 0, 1, \dots, m$:

$$\sum_{n=0}^m 2^n a_{2^{n+1}} \leq \sum_{k=1}^{2^{m+1}-1} a_k \leq \sum_{n=0}^m 2^n a_{2^n}.$$

Thus $\sum a_n$ converges iff $\sum 2^n a_{2^n}$ converges.

Exercise 6.5

Suppose $f : [0, \infty) \rightarrow \mathbb{R}$ is nonnegative and decreasing. Show that

$$\sum_{n=1}^\infty f(n) \text{ converges} \iff \sum_{k=0}^\infty 2^k f(2^k) \text{ converges}.$$

Solution. Let $a_n = f(n)$. Then (a_n) is decreasing and nonnegative. By the Cauchy Condensation Test (Exercise 6.4),

$$\sum_{n=1}^{\infty} a_n \text{ converges} \iff \sum_{k=0}^{\infty} 2^k a_{2^k} \text{ converges.}$$

Substituting $a_n = f(n)$ yields the desired result:

$$\sum_{n=1}^{\infty} f(n) \text{ converges} \iff \sum_{k=0}^{\infty} 2^k f(2^k) \text{ converges.}$$

Exercise 6.6

Show that

$$\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p}$$

converges iff $p > 1$.

Solution. Define $a_n = \frac{1}{n(\log n)^p}$. For $n \geq 3$, (a_n) is positive and decreasing. By the condensation test,

$$\sum_{n=2}^{\infty} a_n \text{ converges} \iff \sum_{k=1}^{\infty} 2^k a_{2^k} \text{ converges.}$$

Now

$$2^k a_{2^k} = \frac{2^k}{2^k (\log 2^k)^p} = \frac{1}{(k \log 2)^p}.$$

Hence the condensed series is

$$\sum_{k=1}^{\infty} \frac{1}{(k \log 2)^p} = \frac{1}{(\log 2)^p} \sum_{k=1}^{\infty} \frac{1}{k^p}.$$

The p -series $\sum 1/k^p$ converges iff $p > 1$. Therefore the given series converges iff $p > 1$.

Exercise 6.1

Suppose $f : [0, 1) \rightarrow [0, 1)$ is nonnegative, continuous, and monotone decreasing. Let $N, M \in \mathbb{Z}$.

a) Show that

$$\sum_{n=N}^M f(n) \geq \int_N^{M+1} f(x) dx.$$

b) Show that

$$\sum_{n=N+1}^M f(n) \leq \int_N^M f(x) dx.$$

c) Deduce

$$\int_N^{M+1} f(x) dx \leq \sum_{n=N}^M f(n) \leq f(N) + \int_N^M f(x) dx.$$

d) Show that $\sum_{n=N}^{\infty} f(n)$ converges iff the improper integral $\int_N^{\infty} f(x) dx$ exists.

e) Conclude that $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

Solution. (a) On $[n, n+1]$, $f(x) \leq f(n)$, so

$$\int_n^{n+1} f(x) dx \leq f(n).$$

Summing yields

$$\int_N^{M+1} f(x) dx \leq \sum_{n=N}^M f(n).$$

(b) On $[n, n+1]$, $f(x) \geq f(n+1)$, so

$$\int_n^{n+1} f(x) dx \geq f(n+1).$$

Summing gives

$$\int_N^M f(x) dx \geq \sum_{n=N+1}^M f(n).$$

(c) Combining (a) and (b):

$$\int_N^{M+1} f(x) dx \leq \sum_{n=N}^M f(n) \leq f(N) + \int_N^M f(x) dx.$$

(d) The error term between the sum and integral is at most $f(N)$, independent of M . Hence $\sum f(n)$ converges iff $\int f(x)$ converges.

(e) For $f(x) = 1/x$, $\int_1^M \frac{1}{x} dx = \log M \rightarrow \infty$. So the harmonic series diverges.

Exercise 6.2

Show that

$$\int_1^n \log x dx \leq \sum_{k=1}^n \log k \leq \int_1^n \log x dx + \log n.$$

Deduce that

$$\log(n!) \sim n \log n - n \quad (n \rightarrow \infty).$$

Solution. On each interval $[k, k+1]$, we have

$$\int_k^{k+1} \log x dx \leq \log(k+1).$$

Summing over $k = 1, \dots, n-1$ gives

$$\int_1^n \log x dx \leq \sum_{k=2}^n \log k.$$

Adding $\log 1 = 0$ gives

$$\int_1^n \log x dx \leq \sum_{k=1}^n \log k.$$

Similarly, one checks that

$$\sum_{k=1}^n \log k \leq \int_1^n \log x \, dx + \log n.$$

Thus

$$n \log n - n + 1 \leq \log(n!) \leq n \log n - n + 1 + \log n.$$

Dividing through by $n \log n - n$ and letting $n \rightarrow \infty$ shows

$$\frac{\log(n!)}{n \log n - n} \rightarrow 1,$$

i.e.

$$\log(n!) \sim n \log n - n.$$

Exercise 6.3

Let $f : [a, b] \rightarrow \mathbb{R}^m$ be bounded.

- Define what it means for f to be Riemann integrable.
- Show that if each coordinate function $f_i : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable, then f is integrable and

$$\int_a^b f(x) \, dx = \left(\int_a^b f_1(x) \, dx, \dots, \int_a^b f_m(x) \, dx \right).$$

- Conversely, show that if f is integrable then each coordinate f_i is integrable and the same equality holds.

Solution. (a) f is Riemann integrable if there exists $I \in \mathbb{R}^m$ such that for every sequence of tagged partitions with mesh $\rightarrow 0$, $R(f, P_n, \tau_n) \rightarrow I$.

- Writing $f = (f_1, \dots, f_m)$, we have

$$R(f, P, \tau) = (R(f_1, P, \tau), \dots, R(f_m, P, \tau)).$$

If each f_i is integrable, then each component converges to $\int f_i$. Hence f is integrable and

$$\int_a^b f(x) \, dx = \left(\int_a^b f_1(x) \, dx, \dots, \int_a^b f_m(x) \, dx \right).$$

- Conversely, if f is integrable with integral $I = (I_1, \dots, I_m)$, then $R(f, P_n, \tau_n) \rightarrow I$ implies $R(f_i, P_n, \tau_n) \rightarrow I_i$. Thus each f_i is integrable and $\int f_i = I_i$.

Exercise 6.4

Suppose $f, g : [a, b] \rightarrow \mathbb{R}^m$ are Riemann integrable and $\alpha, \beta \in \mathbb{R}$.

- Show that $\alpha f + \beta g$ is integrable and

$$\int_a^b (\alpha f(x) + \beta g(x)) \, dx = \alpha \int_a^b f(x) \, dx + \beta \int_a^b g(x) \, dx.$$

- Show that

$$\left\| \int_a^b f(x) \, dx \right\| \leq \int_a^b \|f(x)\| \, dx.$$

Solution. (a) For a tagged partition (P, τ) ,

$$R(\alpha f + \beta g, P, \tau) = \alpha R(f, P, \tau) + \beta R(g, P, \tau).$$

As $\text{mesh}(P) \rightarrow 0$, these sums converge to the corresponding integrals. Hence

$$\int_a^b (\alpha f + \beta g) = \alpha \int_a^b f + \beta \int_a^b g.$$

(b) Again, for (P, τ) ,

$$\|R(f, P, \tau)\| = \left\| \sum f(t_j)(x_j - x_{j-1}) \right\| \leq \sum \|f(t_j)\|(x_j - x_{j-1}) = R(\|f\|, P, \tau).$$

Passing to the limit as $\text{mesh}(P) \rightarrow 0$ gives

$$\left\| \int_a^b f(x) dx \right\| \leq \int_a^b \|f(x)\| dx.$$

Exercise 6.5 (Optional)

Let $f, g : [a, b] \rightarrow \mathbb{R}^m$ be Riemann integrable.

a) Define the inner product

$$\langle f, g \rangle = \int_a^b f(x) \cdot g(x) dx,$$

and the corresponding norm

$$\|f\|_2 = \left(\int_a^b \|f(x)\|^2 dx \right)^{1/2}.$$

b) Show the Cauchy–Schwarz inequality:

$$|\langle f, g \rangle| \leq \|f\|_2 \|g\|_2.$$

c) Deduce the Minkowski inequality:

$$\|f + g\|_2 \leq \|f\|_2 + \|g\|_2.$$

Solution. (a) The inner product is defined by

$$\langle f, g \rangle = \int_a^b f(x) \cdot g(x) dx,$$

with induced norm

$$\|f\|_2 = (\langle f, f \rangle)^{1/2} = \left(\int_a^b \|f(x)\|^2 dx \right)^{1/2}.$$

(b) Pointwise inequality: $|f(x) \cdot g(x)| \leq \|f(x)\| \|g(x)\|$. Integrating gives

$$|\langle f, g \rangle| \leq \int_a^b \|f(x)\| \|g(x)\| dx.$$

Applying Hölder ($p = q = 2$) yields

$$\int_a^b \|f(x)\| \|g(x)\| \leq \|f\|_2 \|g\|_2.$$

Thus

$$|\langle f, g \rangle| \leq \|f\|_2 \|g\|_2.$$

(c) Expanding,

$$\|f + g\|_2^2 = \|f\|_2^2 + 2\langle f, g \rangle + \|g\|_2^2.$$

By (b), $2\langle f, g \rangle \leq 2\|f\|_2 \|g\|_2$, so

$$\|f + g\|_2^2 \leq (\|f\|_2 + \|g\|_2)^2.$$

Taking square roots gives Minkowski:

$$\|f + g\|_2 \leq \|f\|_2 + \|g\|_2.$$

Exercise 6.3

Suppose $f : [a, b] \rightarrow \mathbb{R}^m$ is integrable.

a) Show that if $x, y \in [a, b]$:

$$|\|f(x)\| - \|f(y)\|| \leq \sum_{l=1}^m |f^l(x) - f^l(y)|.$$

b) Deduce that if I is any subinterval of $[a, b]$:

$$\sup_{x \in I} \|f(x)\| - \inf_{y \in I} \|f(y)\| \leq \sum_{l=1}^m \left(\sup_{x \in I} f^l(x) - \inf_{y \in I} f^l(y) \right).$$

c) Prove that $\|f\| : [a, b] \rightarrow \mathbb{R}$ is integrable.

Solution. (a) By the general inequality $|\|u\| - \|v\|| \leq \|u - v\|$,

$$|\|f(x)\| - \|f(y)\|| \leq \|f(x) - f(y)\|.$$

But

$$\|f(x) - f(y)\| \leq \sum_{l=1}^m |f^l(x) - f^l(y)|.$$

Hence the claim.

(b) Taking supremum over $x \in I$ and infimum over $y \in I$,

$$\sup_{x \in I} \|f(x)\| - \inf_{y \in I} \|f(y)\| \leq \sum_{l=1}^m \left(\sup_{x \in I} f^l(x) - \inf_{y \in I} f^l(y) \right).$$

(c) For any partition $P = (x_0, \dots, x_k)$,

$$U(\|f\|, P) - L(\|f\|, P) = \sum_{j=1}^k \left(\sup_{x \in I_j} \|f(x)\| - \inf_{y \in I_j} \|f(y)\| \right) \Delta x_j.$$

By (b), this is at most

$$\sum_{l=1}^m (U(f^l, P) - L(f^l, P)).$$

Since each f^l is integrable, the right-hand side can be made $< \varepsilon$ for sufficiently fine partitions. Hence $\|f\|$ is integrable.

Exercise 6.4

Let $f : [a, b] \rightarrow \mathbb{R}^m$ be integrable, and write

$$f(t) = e_1 f^1(t) + e_2 f^2(t) + \cdots + e_m f^m(t),$$

for coordinate functions $f^i : [a, b] \rightarrow \mathbb{R}$.

a) For a tagged partition (P, τ) , show that

$$R(f, P, \tau) = e_1 R(f^1, P, \tau) + \cdots + e_m R(f^m, P, \tau).$$

b) Show that if $\{(P_\ell, \tau_\ell)\}$ is a sequence of tagged partitions with mesh $\|P_\ell\| \rightarrow 0$, then

$$R(f, P_\ell, \tau_\ell) \rightarrow \int_a^b f(x) dx.$$

c) Show that for a tagged partition (P, τ) ,

$$\|R(f, P, \tau)\| \leq R(\|f\|, P, \tau).$$

d) Deduce that

$$\left\| \int_a^b f(t) dt \right\| \leq \int_a^b \|f(t)\| dt.$$

Solution. (a) Writing $f(t_j) = (f^1(t_j), \dots, f^m(t_j))$, we have

$$R(f, P, \tau) = \sum_{j=1}^k f(t_j) \Delta x_j = \sum_{j=1}^k \sum_{i=1}^m e_i f^i(t_j) \Delta x_j = \sum_{i=1}^m e_i R(f^i, P, \tau).$$

(b) Since each f^i is integrable, $R(f^i, P_\ell, \tau_\ell) \rightarrow \int_a^b f^i(x) dx$. Thus

$$R(f, P_\ell, \tau_\ell) \rightarrow \sum_{i=1}^m e_i \int_a^b f^i(x) dx = \int_a^b f(x) dx.$$

(c) By the triangle inequality,

$$\|R(f, P, \tau)\| = \left\| \sum_{j=1}^k f(t_j) \Delta x_j \right\| \leq \sum_{j=1}^k \|f(t_j)\| \Delta x_j = R(\|f\|, P, \tau).$$

(d) For a refining sequence of partitions, (c) gives

$$\|R(f, P_\ell, \tau_\ell)\| \leq R(\|f\|, P_\ell, \tau_\ell).$$

Taking limits as $\ell \rightarrow \infty$ yields

$$\left\| \int_a^b f(t) dt \right\| \leq \int_a^b \|f(t)\| dt.$$

Definition (Tagged partition). A *partition* of $[a, b]$ is a finite ordered set

$$P = \{x_0, x_1, \dots, x_k\}, \quad a = x_0 < x_1 < \dots < x_k = b.$$

Its *mesh* is

$$\|P\| = \max_{1 \leq j \leq k} (x_j - x_{j-1}).$$

A *tagged partition* is a partition $P = (x_0, \dots, x_k)$ together with points (tags) $t_j \in [x_{j-1}, x_j]$ for $j = 1, \dots, k$.

For a bounded function $f : [a, b] \rightarrow \mathbb{R}^m$, the *Riemann sum* with respect to (P, τ) is

$$R(f, P, \tau) = \sum_{j=1}^k f(t_j) (x_j - x_{j-1}).$$

Exercise 6.5 (*)

Suppose $f, g, h : [a, b] \rightarrow \mathbb{R}^m$ are integrable. Define

$$(f, g) := \int_a^b \langle f(t), g(t) \rangle dt, \quad \|f\|_{L^2} = \sqrt{(f, f)}.$$

a) Show that

$$(f, g) = (g, f), \quad (f + h, g) = (f, g) + (h, g), \quad (\lambda f, g) = \lambda(f, g).$$

b) For $t \in \mathbb{R}$, show that

$$\|f + tg\|_{L^2}^2 = \|f\|_{L^2}^2 + 2t(f, g) + t^2\|g\|_{L^2}^2 \geq 0.$$

c) Deduce the *Cauchy-Schwarz inequality*:

$$|(f, g)| \leq \|f\|_{L^2} \|g\|_{L^2}.$$

When does equality hold?

d) Deduce the *Minkowski inequality*:

$$\|f + g\|_{L^2} \leq \|f\|_{L^2} + \|g\|_{L^2}.$$

Solution. (a) Symmetry: $(f, g) = (g, f)$ since dot product is symmetric. Linearity follows from properties of the integral:

$$(f + h, g) = (f, g) + (h, g), \quad (\lambda f, g) = \lambda(f, g).$$

(b) Expanding,

$$\|f + tg\|_{L^2}^2 = (f + tg, f + tg) = (f, f) + 2t(f, g) + t^2(g, g) = \|f\|_{L^2}^2 + 2t(f, g) + t^2\|g\|_{L^2}^2 \geq 0.$$

(c) Since the quadratic in t is nonnegative, its discriminant is nonpositive:

$$(2(f, g))^2 - 4\|f\|_{L^2}^2\|g\|_{L^2}^2 \leq 0.$$

Hence

$$|(f, g)| \leq \|f\|_{L^2} \|g\|_{L^2}.$$

Equality holds when f and g are linearly dependent almost everywhere.

(d) Expanding,

$$\|f + g\|_{L^2}^2 = \|f\|_{L^2}^2 + 2(f, g) + \|g\|_{L^2}^2.$$

By (c),

$$2(f, g) \leq 2\|f\|_{L^2} \|g\|_{L^2}.$$

Thus

$$\|f + g\|_{L^2}^2 \leq (\|f\|_{L^2} + \|g\|_{L^2})^2,$$

so

$$\|f + g\|_{L^2} \leq \|f\|_{L^2} + \|g\|_{L^2}.$$

Exercise 7.1

Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is given by $f(x) = x$.

Solution. For any $p \in \mathbb{R}^n$,

$$f(p + h) - f(p) = h.$$

Taking $L = \iota$ (the identity map),

$$\frac{\|f(p + h) - f(p) - Lh\|}{\|h\|} = \frac{\|h - \iota h\|}{\|h\|} = 0.$$

Hence f is differentiable at every p with derivative $Df(p) = \iota$.

Exercise 7.2

Show that the map $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$f\begin{pmatrix} x \\ y \end{pmatrix} = x^2 + y^2$$

is differentiable at all $p = (\xi, \eta) \in \mathbb{R}^2$ with Jacobian

$$Df(p) = (2\xi \ 2\eta).$$

Solution. The partial derivatives are

$$\frac{\partial f}{\partial x}(x, y) = 2x, \quad \frac{\partial f}{\partial y}(x, y) = 2y.$$

So the gradient row vector is

$$Df(x, y) = (2x \ 2y).$$

At $p = (\xi, \eta)$,

$$Df(p) = (2\xi \ 2\eta).$$

Since f is a polynomial, it is smooth and thus differentiable everywhere.

Exercise 7.3

One might hope to compute the differential via

$$\lim_{x \rightarrow p} \frac{f(x) - f(p)}{\|x - p\|}.$$

Show this limit need not exist, even if f is differentiable.

Solution. Take $f(x) = x : \mathbb{R}^n \rightarrow \mathbb{R}^n$, differentiable everywhere with derivative the identity. At $p = 0$,

$$\frac{f(x) - f(0)}{\|x\|} = \frac{x}{\|x\|}.$$

As $x \rightarrow 0$, this depends on the direction: along e_1 it tends to e_1 , along e_2 to e_2 . Thus the limit does not exist, though f is differentiable at 0.

Exercise 7.4

Suppose $\Omega \subset \mathbb{R}^n$ is open and $f, g : \Omega \rightarrow \mathbb{R}^m$ are differentiable at $p \in \Omega$. Show that $h = f + g$ is differentiable at p and

$$Dh(p) = Df(p) + Dg(p).$$

Solution. For $h = f + g$,

$$h(p + h) - h(p) = (f(p + h) - f(p)) + (g(p + h) - g(p)).$$

Subtract $(Df(p) + Dg(p))h$ and divide by $\|h\|$. Each term tends to 0 by differentiability of f and g . Hence h is differentiable and $Dh(p) = Df(p) + Dg(p)$.

Exercise 7.5

Suppose $\Omega, \Omega' \subset \mathbb{R}^n$ are open, $g : \Omega \rightarrow \Omega'$, $f : \Omega' \rightarrow \Omega$ with $f \circ g = \text{id}_\Omega$, $g \circ f = \text{id}_{\Omega'}$. Assume g is differentiable at $p \in \Omega$ and f is differentiable at $g(p) \in \Omega'$. Show

$$Df(g(p)) = (Dg(p))^{-1}.$$

Solution. From $f \circ g = \text{id}_\Omega$, the chain rule at p gives

$$Df(g(p)) \cdot Dg(p) = I.$$

From $g \circ f = \text{id}_{\Omega'}$, differentiating at $g(p)$ gives

$$Dg(p) \cdot Df(g(p)) = I.$$

Thus $Df(g(p))$ is both a left- and right-inverse of $Dg(p)$, hence

$$Df(g(p)) = (Dg(p))^{-1}.$$

Exercise 7.6 (*)

(a) Show that the map $P : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by

$$P\begin{pmatrix} x \\ y \end{pmatrix} = xy$$

is differentiable at $p = (\xi, \eta)$ with Jacobian

$$DP(p) = (\eta \ \xi).$$

Solution. Partial derivatives:

$$\frac{\partial P}{\partial x}(x, y) = y, \quad \frac{\partial P}{\partial y}(x, y) = x.$$

At (ξ, η) ,

$$DP(\xi, \eta) = (\eta \ \xi).$$

Since P is a polynomial, it is differentiable everywhere.

(b) Suppose $f, g : \mathbb{R}^n \rightarrow \mathbb{R}$ are differentiable at q . Show that

$$Q(z) = \begin{pmatrix} f(z) \\ g(z) \end{pmatrix}$$

is differentiable with

$$DQ(q) = \begin{pmatrix} Df(q) \\ Dg(q) \end{pmatrix}.$$

Solution. Q has components $Q_1 = f$, $Q_2 = g$. Since f, g are differentiable at q , Q is differentiable with Jacobian

$$DQ(q) = \begin{pmatrix} Df(q) \\ Dg(q) \end{pmatrix}.$$

(c) Show that $F(z) = f(z)g(z)$ is differentiable with

$$DF(q) = g(q)Df(q) + f(q)Dg(q).$$

Solution. Write $F = P \circ Q$, where $P(x, y) = xy$, $Q(z) = (f(z), g(z))$. By the chain rule,

$$DF(q) = DP(Q(q)) \cdot DQ(q).$$

Now $DP(Q(q)) = (g(q) \ f(q))$, and $DQ(q) = \begin{pmatrix} Df(q) \\ Dg(q) \end{pmatrix}$. So

$$DF(q) = (g(q) \ f(q)) \begin{pmatrix} Df(q) \\ Dg(q) \end{pmatrix} = g(q)Df(q) + f(q)Dg(q).$$

Exercise 8.1

Show that $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is everywhere differentiable and compute the differential.

a) $f(x, y) = x^2 + y^2 - x - xy.$

$$Df(x, y) = (2x - 1 - y, 2y - x).$$

b) $f(x, y) = \frac{1}{\sqrt{1 + x^2 + y^2}}.$

$$Df(x, y) = \left(-\frac{x}{(1 + x^2 + y^2)^{3/2}}, -\frac{y}{(1 + x^2 + y^2)^{3/2}} \right).$$

c) $f(x, y) = x^5 y^2.$

$$Df(x, y) = (5x^4 y^2, 2x^5 y).$$

Exercise 8.2

Find the minimum of

$$f(x, y, z) = x^4(y^2 + x^2) + z^2 - 4z.$$

Solution. $f_x = 2x^3(2y^2 + 3x^2)$, $f_y = 2x^4 y$, $f_z = 2z - 4$. Solving $\nabla f = 0$ gives $x = 0, z = 2$, y arbitrary.

Thus the critical set is $\{(0, y, 2) : y \in \mathbb{R}\}$. At these points,

$$f(0, y, 2) = -4.$$

Hence the minimum value is -4 , attained along the line $\{(0, y, 2) : y \in \mathbb{R}\}$.

Exercise 8.3

Suppose A is a symmetric $n \times n$ matrix and $f(x) = x^t A x$.

a) Show that f is differentiable with

$$Df(p) = 2p^t A.$$

Solution. $\nabla f(x) = (A + A^t)x = 2Ax$ since A is symmetric. Thus $Df(p) = (2Ap)^t = 2p^t A$.

b) Find Hess $f(p)$.

Solution. $\nabla f(x) = 2Ax$, so the derivative is constant:

$$\text{Hess } f(p) = 2A.$$

Exercise 8.4 (*)

Consider

$$f(x, y) = \begin{cases} \frac{xy^3 - x^3y}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

a) Show that

$$D_1f(x, y) = \begin{cases} \frac{y^3 - 3x^2y}{x^2 + y^2} - \frac{2x(xy^3 - x^3y)}{(x^2 + y^2)^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0), \end{cases}$$

and

$$D_2f(x, y) = \begin{cases} \frac{3y^2x - x^3}{x^2 + y^2} - \frac{2y(xy^3 - x^3y)}{(x^2 + y^2)^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

b) Show that D_1f and D_2f are continuous at $(0, 0)$.

Solution. For $(x, y) \neq (0, 0)$ the formulas follow by quotient rule. At $(0, 0)$ the limits defining partial derivatives equal 0. In polar coordinates $(x, y) = (r \cos \theta, r \sin \theta)$, both D_1f, D_2f are $O(r)$ as $r \rightarrow 0$, hence tend to 0. Thus both are continuous at $(0, 0)$.

Exercise 8.4 (b,c)

b) Along $te_2 = (0, t)$,

$$D_1f(0, t) = t \Rightarrow \frac{D_1f(0, t) - D_1f(0, 0)}{t} = 1.$$

Along $te_1 = (t, 0)$,

$$D_2f(t, 0) = -t \Rightarrow \frac{D_2f(t, 0) - D_2f(0, 0)}{t} = -1.$$

c) Hence

$$D_2D_1f(0) = 1, \quad D_1D_2f(0) = -1.$$

So both exist, but are not equal.

Exercise 8.5 (*)

a) By induction on m , prove the multinomial theorem:

$$(x_1 + \cdots + x_m)^j = \sum_{n_1 + \cdots + n_m = j} \binom{j}{n_1, \dots, n_m} x_1^{n_1} \cdots x_m^{n_m}.$$

Solution. For $m = 1$ trivial; for $m = 2$ this is the binomial theorem. Inductive step: write

$$(x_1 + \cdots + x_m)^j = \sum_{k=0}^j \binom{j}{k} (x_1 + \cdots + x_{m-1})^{j-k} x_m^k.$$

Apply induction to $(x_1 + \cdots + x_{m-1})^{j-k}$, relabel $n_m = k$, and simplify coefficients to obtain

$$\binom{j}{n_1, \dots, n_m} = \frac{j!}{n_1! \cdots n_m!}.$$

b) Expanding

$$(x_1 + \cdots + x_m)^j = (x_1 + \cdots + x_m)(x_1 + \cdots + x_m)^{j-1},$$

the coefficient of $x_1^{n_1} \cdots x_m^{n_m}$ on the LHS is $\binom{j}{n_1, \dots, n_m}$. On the RHS it comes from choosing x_i from the first factor, giving

$$\binom{j-1}{n_1, \dots, n_i-1, \dots, n_m}.$$

Summing over $i = 1, \dots, m$ gives the recursive identity.

Exercise 8.6

Let $\Omega = \{(x, y)^t \in \mathbb{R}^2 : x > 0\}$ and

$$f(x, y) = \begin{pmatrix} x \sin y \\ x \cos y \end{pmatrix}.$$

a) Compute derivatives:

$$Df(\xi, \eta) = \begin{pmatrix} \sin \eta & \xi \cos \eta \\ \cos \eta & -\xi \sin \eta \end{pmatrix}.$$

Thus f is differentiable everywhere.

b) $\det Df(\xi, \eta) = -\xi \neq 0$ for $(\xi, \eta) \in \Omega$, hence invertible.

c) f is not injective since $f(x, y) = f(x, y + 2\pi)$. So invertibility of the derivative does not imply global injectivity. This illustrates the local nature of the Inverse Function Theorem.

Exercise 8.7

a) Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuously differentiable near 0 and $f'(0) = 0$. We need an example showing that f may still be bijective.

Solution. Consider

$$f(x) = x^3.$$

Then $f'(x) = 3x^2$, so $f'(0) = 0$. Nevertheless, f is strictly increasing on $\mathbb{R}$, hence bijective with inverse

$$f^{-1}(y) = \sqrt[3]{y}.$$

Thus, even if the derivative at the origin vanishes, f may still be bijective in one dimension.

b) Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is bijective, differentiable at the origin, and $\det Df(0) = 0$. Show that f^{-1} is not differentiable at $f(0)$.

Solution. Assume for contradiction that f^{-1} is differentiable at $f(0)$. Define the identity map $\iota : \mathbb{R}^n \rightarrow \mathbb{R}^n$ by $\iota(x) = x$. Note that $\iota = f^{-1} \circ f$.

Differentiating at 0 and applying the chain rule gives

$$D\iota(0) = D(f^{-1})(f(0)) \cdot Df(0).$$

But $D\iota(0) = I_n$, the identity matrix. Thus

$$I_n = D(f^{-1})(f(0)) \cdot Df(0).$$

This equation says that $Df(0)$ must have a right inverse, which would imply that $Df(0)$ is invertible. But $\det Df(0) = 0$, so $Df(0)$ is singular and not invertible. This is a contradiction.

Therefore, f^{-1} cannot be differentiable at $f(0)$.

Exercise 9.1

Let $\gamma : [0, 2\pi] \rightarrow \mathbb{R}^2$ be given by

$$\gamma(\theta) = \begin{pmatrix} \theta - \sin \theta \\ 1 - \cos \theta \end{pmatrix}.$$

Definition (Parametrised curve). A *parametrised curve* in $\mathbb{R}^2$ is a map $\gamma : [a, b] \rightarrow \mathbb{R}^2$. We say γ is of class C^1 if its coordinate functions are continuously differentiable.

Definition (Arc length). For a C^1 curve $\gamma : [a, b] \rightarrow \mathbb{R}^2$, the *arc length* is

$$L(\gamma) = \int_a^b \|\gamma'(t)\| dt,$$

where $\|\cdot\|$ is the Euclidean norm in $\mathbb{R}^2$.

Definition (Cycloid). The *cycloid* is the path traced by a fixed point on the circumference of a circle of radius 1 rolling along the x -axis without slipping. Its standard parametrisation is

$$\gamma(\theta) = (\theta - \sin \theta, 1 - \cos \theta), \quad 0 \leq \theta \leq 2\pi.$$

a) *Sketch.*

At $\theta = 0$, $\gamma(0) = (0, 0)$. At $\theta = \pi$, $\gamma(\pi) = (\pi, 2)$. At $\theta = 2\pi$, $\gamma(2\pi) = (2\pi, 0)$.

Hence γ describes a single arch of the cycloid: it starts at the origin, rises to height 2 above $x = \pi$, and returns to the x -axis at $(2\pi, 0)$. It is symmetric about the vertical line $x = \pi$.

b) *Arc length.*

Differentiate:

$$\gamma'(\theta) = (1 - \cos \theta, \sin \theta).$$

Compute norm:

$$\|\gamma'(\theta)\|^2 = (1 - \cos \theta)^2 + \sin^2 \theta = 2 - 2 \cos \theta = 4 \sin^2\left(\frac{\theta}{2}\right).$$

Thus

$$\|\gamma'(\theta)\| = 2 \sin\left(\frac{\theta}{2}\right), \quad 0 \leq \theta \leq 2\pi.$$

Hence

$$L(\gamma) = \int_0^{2\pi} 2 \sin\left(\frac{\theta}{2}\right) d\theta.$$

Substitute $u = \frac{\theta}{2}$, so $d\theta = 2du$, limits $0 \mapsto 0$, $2\pi \mapsto \pi$:

$$L(\gamma) = 4 \int_0^\pi \sin u du = 4[-\cos u]_0^\pi = 4(1 - (-1)) = 8.$$

$L(\gamma) = 8$

Exercise 9.2

Let $A = [a_1, b_1] \times [a_2, b_2] \subset \mathbb{R}^2$ be a rectangle, and let $\mathcal{P}$ be a partition of A . Suppose $f, g : A \rightarrow \mathbb{R}$ are bounded and $\lambda \geq 0$.

Definition (Upper and Lower Sums). For a bounded function $f : A \rightarrow \mathbb{R}$ and a partition $\mathcal{P}$ of A into rectangles R , the *upper sum* is

$$U(f, \mathcal{P}) = \sum_{R \in \mathcal{P}} \left(\sup_R f \right) |R|,$$

and the *lower sum* is

$$L(f, \mathcal{P}) = \sum_{R \in \mathcal{P}} \left(\inf_R f \right) |R|,$$

where $|R|$ denotes the area of the rectangle R .

Background. These constructions satisfy simple algebraic rules: - taking negatives exchanges suprema and infima, - scaling by a positive constant factors out, - suprema are subadditive, infima are superadditive.

- a) $U(-f, \mathcal{P}) = -L(f, \mathcal{P})$, since $\sup_R(-f) = -\inf_R f$.
- b) $L(-f, \mathcal{P}) = -U(f, \mathcal{P})$, since $\inf_R(-f) = -\sup_R f$.
- c) $U(\lambda f, \mathcal{P}) = \lambda U(f, \mathcal{P})$, since $\sup_R(\lambda f) = \lambda \sup_R f$.
- d) $L(\lambda f, \mathcal{P}) = \lambda L(f, \mathcal{P})$, since $\inf_R(\lambda f) = \lambda \inf_R f$.
- e) $U(f + g, \mathcal{P}) \leq U(f, \mathcal{P}) + U(g, \mathcal{P})$, since $\sup_R(f + g) \leq \sup_R f + \sup_R g$.
- f) $L(f + g, \mathcal{P}) \geq L(f, \mathcal{P}) + L(g, \mathcal{P})$, since $\inf_R(f + g) \geq \inf_R f + \inf_R g$.

Hence the algebraic properties of upper and lower sums hold as claimed.

Exercise 9.3

Let $A = [-1, 1] \times [-1, 1]$ and define

$$f(x, y) = \begin{cases} 1, & x \notin \mathbb{Q}, y = 0, \\ 0, & \text{otherwise.} \end{cases}$$

Background. - The function f takes values only 0 or 1, so it is bounded. - It is nonzero only along the horizontal line segment

$$S = \{(x, 0) : x \in [-1, 1], x \notin \mathbb{Q}\}.$$

- The set S is a subset of a line in $\mathbb{R}^2$, and such sets have *measure zero*. - For the Riemann integral, bounded functions supported on measure-zero sets are integrable with integral 0.

Solution. Since S has measure zero, the contribution of f to any upper sum can be made arbitrarily small. Thus f is Riemann integrable on A , and

$$\int_A f(z) dz = 0.$$

Conclusion. Although f takes the value 1 on uncountably many points (all irrationals on the x -axis between -1 and 1), this set is “too thin” to contribute to the integral. The integral only depends on two-dimensional area, not one-dimensional subsets.

Exercise 9.4

Let $A = [a_1, b_1] \times [a_2, b_2]$ and $|A| = (b_1 - a_1)(b_2 - a_2)$. Suppose $f, g : A \rightarrow \mathbb{R}$ are integrable.

Background. For a bounded function f on A , let

$$m = \inf_{x \in A} f(x), \quad M = \sup_{x \in A} f(x).$$

Then $m \leq f(x) \leq M$ for all $x \in A$. Multiplying by the area $|A|$ and integrating yields inequalities for the integral. Key facts:

- If $f \geq 0$, then $\int_A f \geq 0$.
- If $f \leq g$, then $\int_A f \leq \int_A g$.
- The triangle inequality: $|\int f| \leq \int |f|$.

a) For all $x \in A$, $\inf_A f \leq f(x) \leq \sup_A f$. Integrating:

$$|A| \inf_{x \in A} f(x) \leq \int_A f(x) dx \leq |A| \sup_{x \in A} f(x).$$

b) Applying (a) to both f and $-f$:

$$-|A| \sup_{x \in A} |f(x)| \leq \int_A f(x) dx \leq |A| \sup_{x \in A} |f(x)|.$$

Thus

$$\left| \int_A f(x) dx \right| \leq |A| \sup_{x \in A} |f(x)|.$$

c) If $f(x) \geq 0$, then $\inf_A f \geq 0$. By (a),

$$0 \leq \int_A f(x) dx.$$

d) If $f(x) \leq g(x)$, then $0 \leq g(x) - f(x)$. By (c),

$$0 \leq \int_A (g - f) = \int_A g - \int_A f,$$

so

$$\int_A f \leq \int_A g.$$

e) Since $-f(x) \leq |f(x)|$ and $f(x) \leq |f(x)|$, integrating gives

$$-\int_A |f| \leq \int_A f \leq \int_A |f|.$$

Thus

$$\left| \int_A f(x) dx \right| \leq \int_A |f(x)| dx.$$

Exercise 9.5

Let $A = [a_1, b_1] \times [a_2, b_2]$ and $f : A \rightarrow \mathbb{R}$ continuous. Define

$$F(y) = \int_{a_1}^{b_1} f(x, y) dx, \quad y \in [a_2, b_2].$$

Background. Since f is continuous on the compact rectangle A , it is uniformly continuous. This allows us to control the difference $|f(x, y_1) - f(x, y_2)|$ uniformly in x .

Solution. Fix y_1, y_2 . Then

$$|F(y_1) - F(y_2)| = \left| \int_{a_1}^{b_1} (f(x, y_1) - f(x, y_2)) dx \right| \leq \int_{a_1}^{b_1} |f(x, y_1) - f(x, y_2)| dx.$$

By uniform continuity, for $\varepsilon > 0$ there exists $\delta > 0$ such that if $|y_1 - y_2| < \delta$ then $|f(x, y_1) - f(x, y_2)| < \frac{\varepsilon}{b_1 - a_1}$ for all x . Hence

$$|F(y_1) - F(y_2)| \leq (b_1 - a_1) \frac{\varepsilon}{b_1 - a_1} = \varepsilon.$$

Thus F is continuous.

Exercise 9.6

a) The improper integral

$$\int_1^\infty e^{-u} du = [-e^{-u}]_1^\infty = e^{-1}$$

converges.

b) For $R > 1$,

$$\int_1^R e^{-u^2} du \leq \int_1^R e^{-u} du < \infty,$$

so $\int_1^\infty e^{-u^2} du$ converges. By symmetry the whole integral

$$I = \int_{-\infty}^\infty e^{-u^2} du$$

converges.

c) For $f \geq 0$, with $S_R = [-R, R] \times [-R, R]$ and $B_R(0)$ the disk,

$$\int_{B_R(0)} f \leq \int_{S_R} f \leq \int_{B_{\sqrt{2}R}(0)} f,$$

since $B_R(0) \subset S_R \subset B_{\sqrt{2}R}(0)$.

d) For $f(u, v) = e^{-(u^2+v^2)}$,

$$\iint_{\mathbb{R}^2} e^{-(u^2+v^2)} dudv = \left(\int_{-\infty}^\infty e^{-u^2} du \right)^2 = I^2.$$

In polar coordinates,

$$\iint_{\mathbb{R}^2} e^{-(u^2+v^2)} dudv = \int_0^{2\pi} \int_0^\infty e^{-r^2} r dr d\theta = 2\pi \cdot \frac{1}{2} \int_0^\infty e^{-t} dt = \pi.$$

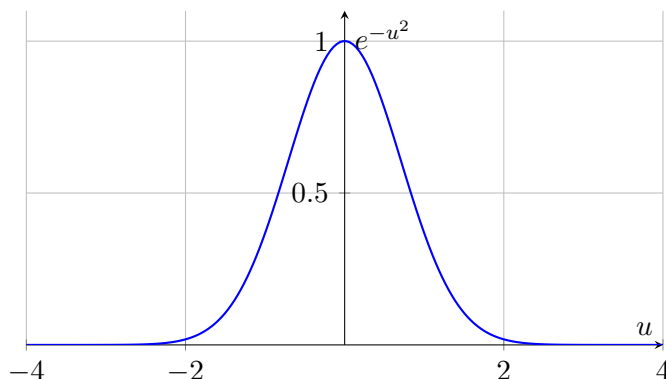
So $I^2 = \pi$.

e) Therefore

$$I = \sqrt{\pi}.$$

Exercise 9.6 (with integrand chart)

The integrand is $f(u) = e^{-u^2}$. It is symmetric, maximised at $u = 0$ with $f(0) = 1$, and decays rapidly to 0 as $|u| \rightarrow \infty$. This rapid decay ensures convergence of the Gaussian integral.



This graph shows the bell-shaped curve of e^{-u^2} . It is clear that the function is bounded and rapidly decays to zero outside a neighbourhood of 0, so the area under the curve converges. Indeed,

$$I = \int_{-\infty}^{\infty} e^{-u^2} du = \sqrt{\pi}.$$

Polar Coordinates. To evaluate

$$I = \int_{-\infty}^{\infty} e^{-u^2} du,$$

we square it:

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-u^2} du \right) \left(\int_{-\infty}^{\infty} e^{-v^2} dv \right) = \iint_{\mathbb{R}^2} e^{-(u^2+v^2)} du dv.$$

Since the integrand depends only on $u^2 + v^2 = r^2$, we switch to polar coordinates:

$$u = r \cos \theta, \quad v = r \sin \theta, \quad du dv = r dr d\theta.$$

Thus

$$I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta = \left(\int_0^{2\pi} d\theta \right) \left(\int_0^{\infty} r e^{-r^2} dr \right).$$

Let $t = r^2$, $dt = 2r dr$:

$$\int_0^{\infty} r e^{-r^2} dr = \frac{1}{2} \int_0^{\infty} e^{-t} dt = \frac{1}{2}.$$

So

$$I^2 = 2\pi \cdot \frac{1}{2} = \pi, \quad I = \sqrt{\pi}.$$

Theorem 1 (Heine–Cantor). *Suppose that $E \subset \mathbb{R}^n$ is compact and $f : E \rightarrow \mathbb{R}^m$ is continuous on E . Then f is uniformly continuous on E .*

Examples

Example 1. Let $f(x) = \sqrt{x}$ on $E = [0, 1]$. Here E is compact and f is continuous. By Heine–Cantor, f is uniformly continuous, even though $f'(x) = \frac{1}{2\sqrt{x}}$ is unbounded near 0.

Example 2. Let $f(x) = \sin(x)$ on $E = [0, 100]$. E is compact, f is continuous, hence uniformly continuous.

Example 3. Let $f(x) = x^2$ on $E = [-5, 5]$. Polynomials are continuous, and since E is compact, f is uniformly continuous.

Example 4. In higher dimensions, let $f(x, y) = \frac{1}{1+x^2+y^2}$ on the closed disk $E = \{(x, y) : x^2 + y^2 \leq 10\}$. The domain is compact and f is continuous, so f is uniformly continuous.

Counterexamples (non-compact domains)

Counterexample 1. Let $f(x) = \sqrt{x}$ on $E = (0, \infty)$. Though f is continuous, it is not uniformly continuous: near 0, small changes in x produce large changes in $\sqrt{x}$. Formally, no single δ works for all ε .

Counterexample 2. Let $f(x) = x^2$ on $E = \mathbb{R}$. Take $x = n$ and $y = n + \frac{1}{n}$. Then $|x - y| = \frac{1}{n} \rightarrow 0$ but $|x^2 - y^2| \approx 2$, so f is not uniformly continuous.

Counterexample 3. Let $f(x) = \sin(x^2)$ on $E = \mathbb{R}$. This function oscillates increasingly rapidly as $x \rightarrow \infty$, so it fails uniform continuity.

Summary Table

Function f	Domain E	Compact?	Continuous?	Uniformly continuous?
$\sqrt{x}$	$[0, 1]$	Yes	Yes	Yes
$\sin(x)$	$[0, 100]$	Yes	Yes	Yes
x^2	$[-5, 5]$	Yes	Yes	Yes
$\frac{1}{1+x^2+y^2}$	Closed disk	Yes	Yes	Yes
$\sqrt{x}$	$(0, \infty)$	No	Yes	No
x^2	$\mathbb{R}$	No	Yes	No
$\sin(x^2)$	$\mathbb{R}$	No	Yes	No

Illustrative Charts

section*Pointwise vs. Uniform Convergence

We visualize functions $f_n(x)$ converging to a limit $f(x)$ inside an ε -band.

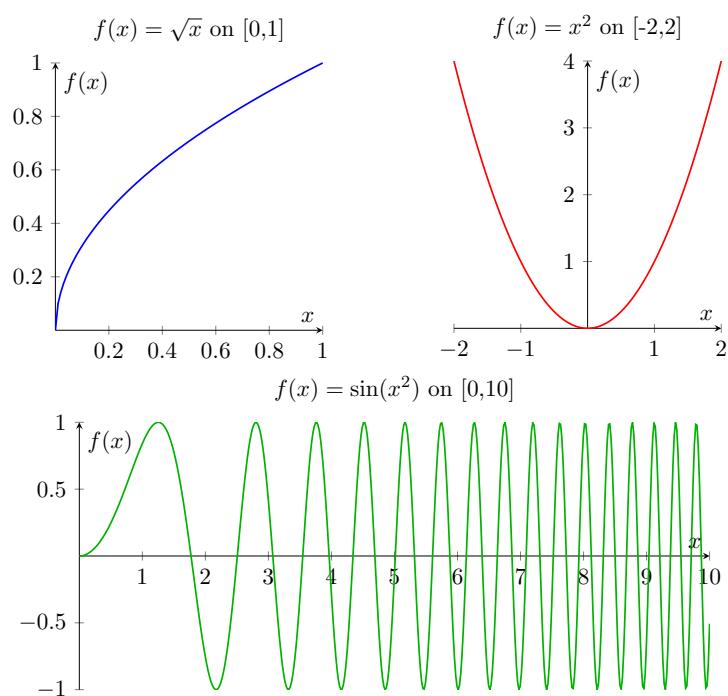
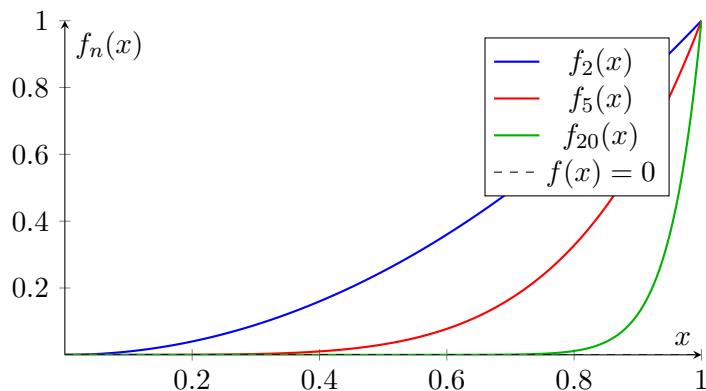


Figure 1: Examples and counterexamples for uniform continuity.

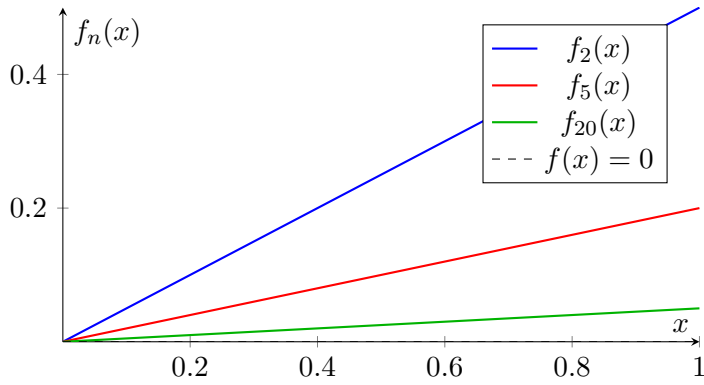
1. Pointwise but not uniform: $f_n(x) = x^n$ on $[0, 1]$

Here $f_n(x) \rightarrow 0$ for $x \in [0, 1)$ and $f_n(1) = 1$, so convergence is pointwise but not uniform.



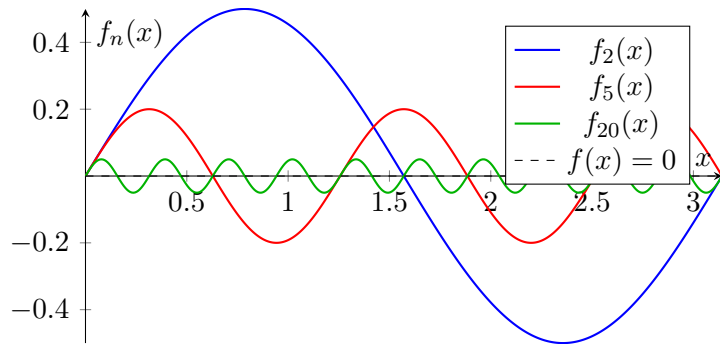
2. Uniform convergence: $f_n(x) = \frac{x}{n}$ on $[0, 1]$

Here $f_n(x) \rightarrow 0$ uniformly, since $\sup_{x \in [0,1]} |f_n(x) - 0| = \frac{1}{n} \rightarrow 0$.



3. Non-uniform oscillations: $f_n(x) = \frac{\sin(nx)}{n}$ on $[0, \pi]$

Pointwise limit is $f(x) = 0$, but uniform convergence fails because oscillations persist near the endpoints.



Exercise 1.1

Consider the meromorphic function

$$F(z) = \frac{1}{1+z^2} \frac{\cos[(\pi-\theta)z]}{2\sin(\pi z)}.$$

a) Poles and residues

The poles occur at $z = \pm i$ and $z = n$, $n \in \mathbb{Z}$. At $z = \pm i$:

$$\text{Res}[F, \pm i] = -\frac{\cosh(\pi - \theta)}{4 \sinh \pi}.$$

At $z = n \in \mathbb{Z}$:

$$\text{Res}[F, n] = \frac{\cos(n\theta)}{2\pi(1+n^2)}.$$

b) The contour $\Gamma^{(N)}$

Define

$$\Gamma_1^{(N)}(t) = \left(N + \frac{1}{2}\right)(1 + it), \quad -1 \leq t \leq 1,$$

$$\Gamma_2^{(N)}(t) = \left(N + \frac{1}{2}\right)(-t + i), \quad -1 \leq t \leq 1,$$

$$\Gamma_3^{(N)}(t) = \left(N + \frac{1}{2}\right)(-1 - it), \quad -1 \leq t \leq 1,$$

$$\Gamma_4^{(N)}(t) = \left(N + \frac{1}{2}\right)(t - i), \quad -1 \leq t \leq 1.$$

Then $\Gamma^{(N)}$ is the closed square contour with vertices $\pm(N + \frac{1}{2}) \pm i(N + \frac{1}{2})$. It encloses the poles at $\pm i$ and all integers n with $|n| \leq N$.

c) Bound on the contour integral

On $\Gamma^{(N)}$, $|z| \sim N$, so $|F(z)| = O(1/N^2)$. Since the contour has length $O(N)$, we obtain

$$\left| \int_{\Gamma^{(N)}} F(z) dz \right| \leq \frac{C}{N},$$

for some constant C independent of N .

d) Residue theorem and Fourier series

By Cauchy's residue theorem,

$$\int_{\Gamma^{(N)}} F(z) dz = 2\pi i \left(\text{Res}[F, i] + \text{Res}[F, -i] + \sum_{n=-N}^N \text{Res}[F, n] \right).$$

Hence

$$\left| \frac{\cosh(\pi - \theta)}{2 \sinh \pi} - \sum_{n=-N}^N \frac{\cos(n\theta)}{1 + n^2} \right| \leq \frac{C}{N}.$$

Letting $N \rightarrow \infty$, we conclude

$$\frac{\cosh(\pi - \theta)}{2 \sinh \pi} = \sum_{n=-\infty}^{\infty} \frac{e^{in\theta}}{1 + n^2}.$$

This gives the Fourier series expansion, uniformly in θ .

Pointwise vs. Uniform Convergence of Functions

We present a collection of classical examples of sequences of functions, with charts illustrating whether convergence is pointwise, uniform, or fails.

—

1. $f_n(x) = x^n$ on $[0, 1]$ (Pointwise but not uniform)

$$f_n(x) \rightarrow f(x) = \begin{cases} 0, & x \in [0, 1), \\ 1, & x = 1. \end{cases}$$

Pointwise but not uniform, since $\|f_n - f\|_{\infty} = 1$.

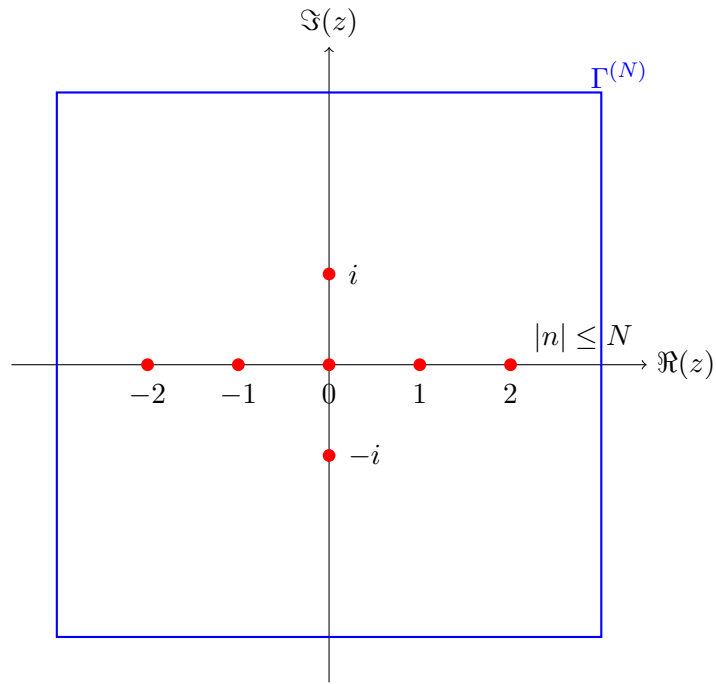
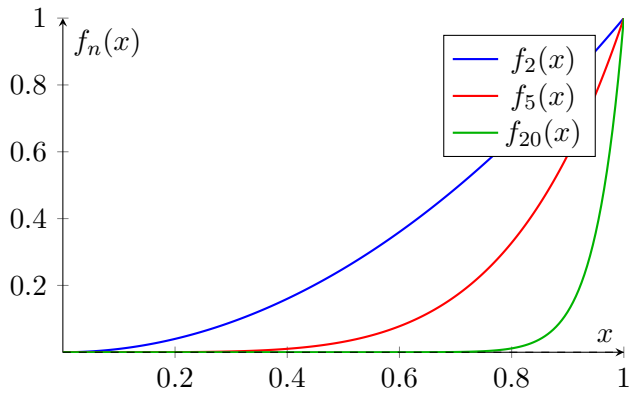
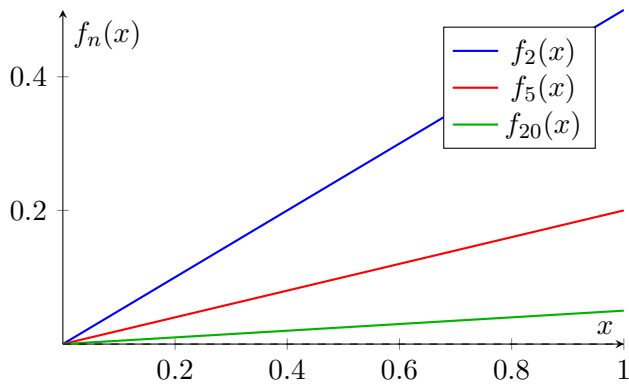


Figure 2: Contour $\Gamma^{(N)}$ enclosing poles at $z = \pm i$ and integers n , $|n| \leq N$.



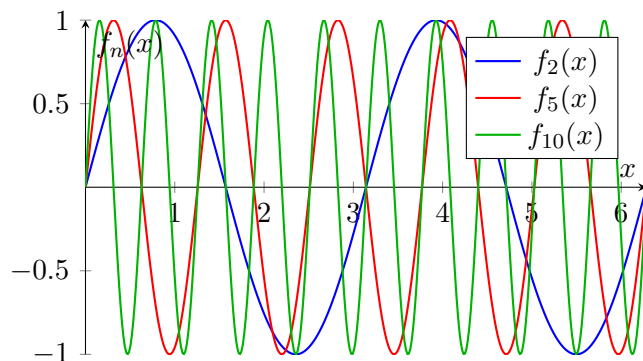
2. $f_n(x) = x/n$ on $[0, 1]$ (Uniform)

$f_n(x) \rightarrow 0$ uniformly, since $\sup_{x \in [0,1]} |f_n(x)| = 1/n \rightarrow 0$.



3. $f_n(x) = \sin(nx)$ on $[0, 2\pi]$ (Oscillatory failure)

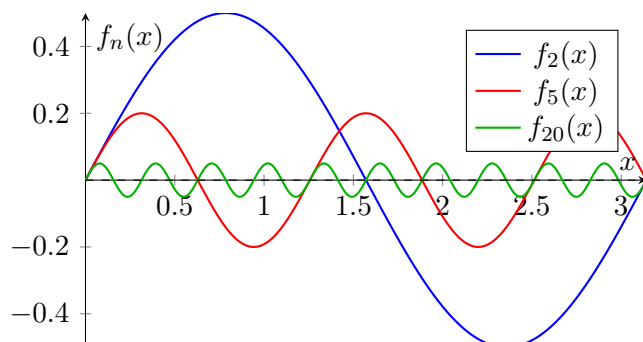
No pointwise limit exists except at some special points. Not uniform: $\|f_n - f_m\|_\infty = 2$ infinitely often.



4. $f_n(x) = \sin(nx)/n$ on $[0, \pi]$ (Uniform with oscillations)

$$|f_n(x)| \leq \frac{1}{n}, \quad \|f_n\|_\infty = 1/n \rightarrow 0.$$

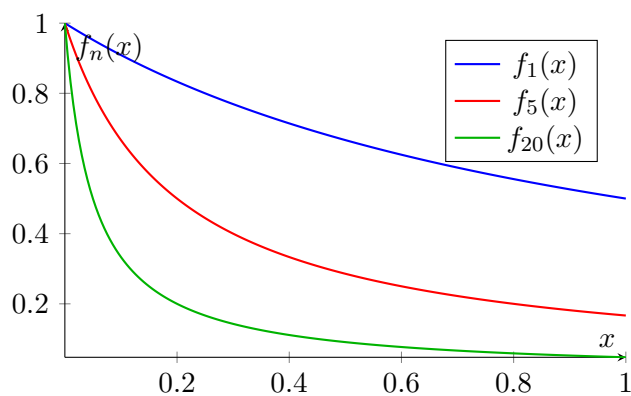
Thus convergence is uniform despite oscillations.



5. $f_n(x) = \frac{1}{1+nx}$ on $[0, 1]$ (Pointwise not uniform)

$$f_n(x) \rightarrow f(x) = \begin{cases} 0, & x > 0, \\ 1, & x = 0. \end{cases}$$

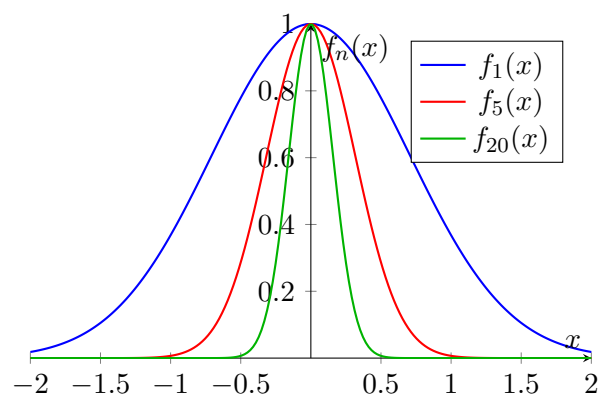
Not uniform because $\|f_n - f\|_\infty = 1$.



6. $f_n(x) = e^{-nx^2}$ on $\mathbb{R}$ (Gaussian bump)

$$f_n(x) \rightarrow f(x) = \begin{cases} 0, & x \neq 0, \\ 1, & x = 0. \end{cases}$$

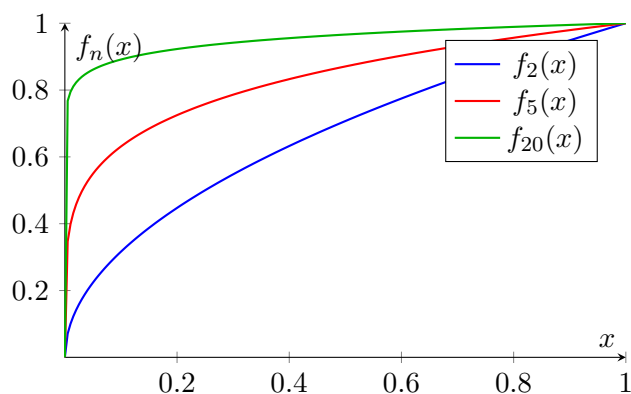
Not uniform, since the peak at 0 prevents uniform convergence.



7. $f_n(x) = x^{1/n}$ on $[0, 1]$ (Discontinuous limit, not uniform)

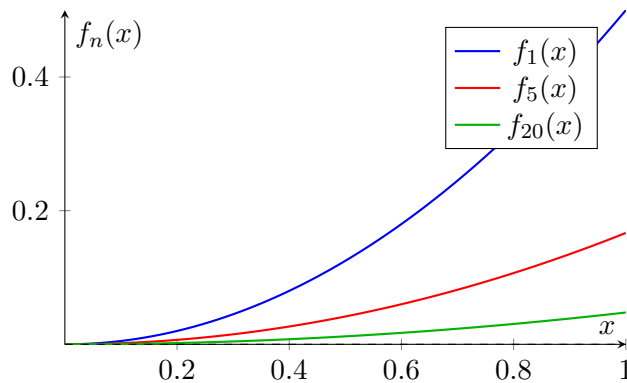
$$f_n(x) \rightarrow f(x) = \begin{cases} 1, & x > 0, \\ 0, & x = 0. \end{cases}$$

Limit function is discontinuous, hence convergence not uniform.



8. $f_n(x) = \frac{x^2}{n+1}$ on $[0, 1]$ (Uniform)

$f_n(x) \rightarrow 0$ uniformly, since $\|f_n\|_\infty = 1/(n+1) \rightarrow 0$.



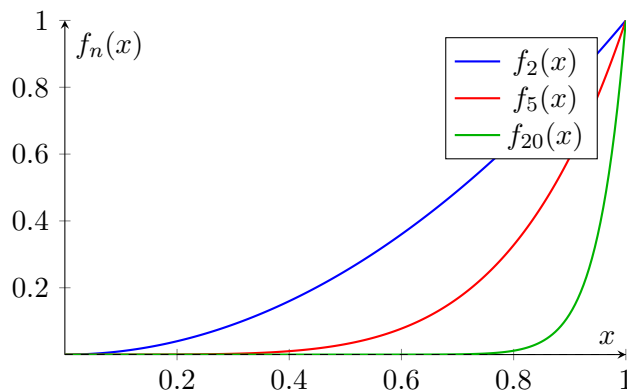
Pointwise vs. Uniform Convergence of Functions

We present a collection of classical examples of sequences of functions, with charts illustrating whether convergence is pointwise, uniform, or fails.

1. $f_n(x) = x^n$ on $[0, 1]$ (Pointwise but not uniform)

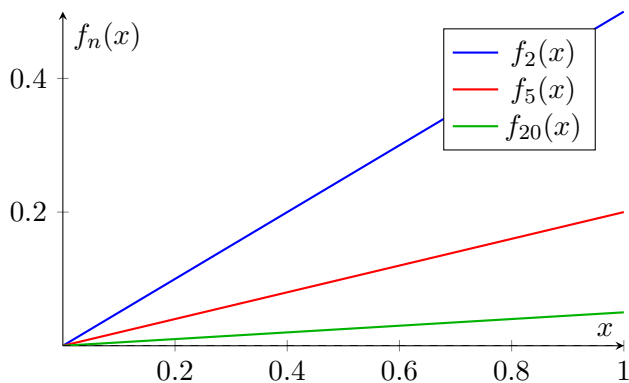
$$f_n(x) \rightarrow f(x) = \begin{cases} 0, & x \in [0, 1), \\ 1, & x = 1. \end{cases}$$

Pointwise but not uniform, since $\|f_n - f\|_\infty = 1$.



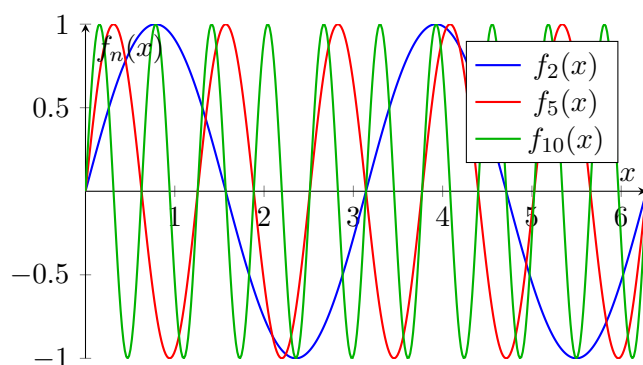
2. $f_n(x) = x/n$ on $[0, 1]$ (Uniform)

$f_n(x) \rightarrow 0$ uniformly, since $\sup_{x \in [0,1]} |f_n(x)| = 1/n \rightarrow 0$.



3. $f_n(x) = \sin(nx)$ on $[0, 2\pi]$ (**Oscillatory failure**)

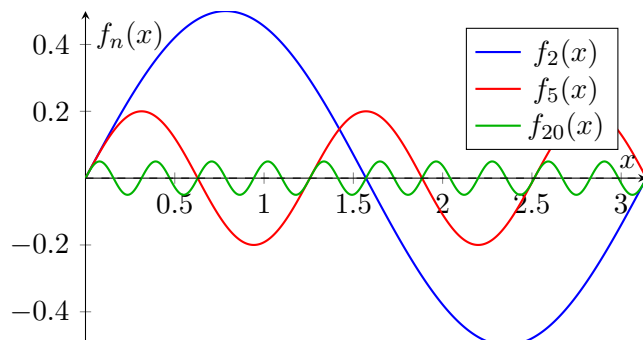
No pointwise limit exists except at some special points. Not uniform: $\|f_n - f_m\|_\infty = 2$ infinitely often.



4. $f_n(x) = \sin(nx)/n$ on $[0, \pi]$ (**Uniform with oscillations**)

$$|f_n(x)| \leq \frac{1}{n}, \quad \|f_n\|_\infty = 1/n \rightarrow 0.$$

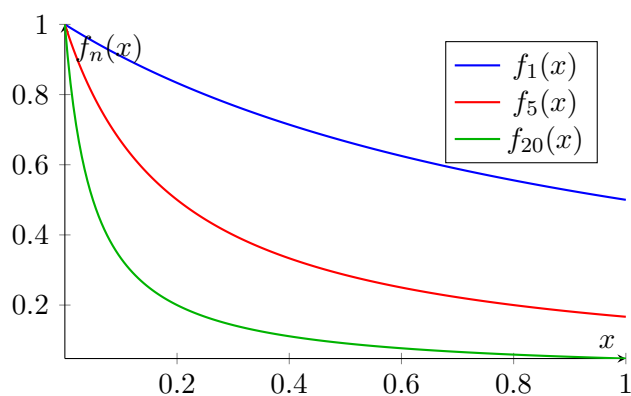
Thus convergence is uniform despite oscillations.



5. $f_n(x) = \frac{1}{1+nx}$ on $[0, 1]$ (**Pointwise not uniform**)

$$f_n(x) \rightarrow f(x) = \begin{cases} 0, & x > 0, \\ 1, & x = 0. \end{cases}$$

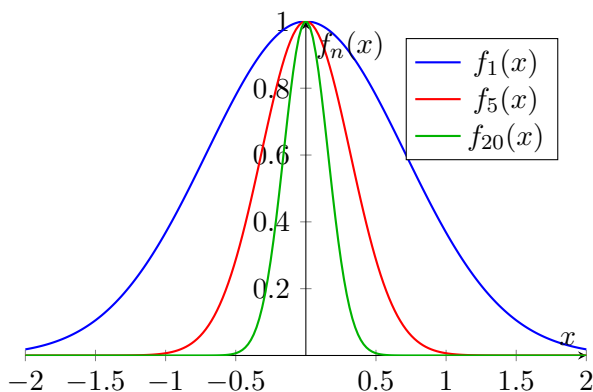
Not uniform because $\|f_n - f\|_\infty = 1$.



6. $f_n(x) = e^{-nx^2}$ on $\mathbb{R}$ (Gaussian bump)

$$f_n(x) \rightarrow f(x) = \begin{cases} 0, & x \neq 0, \\ 1, & x = 0. \end{cases}$$

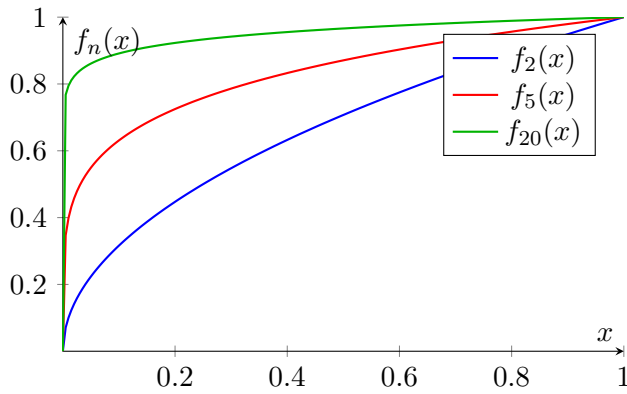
Not uniform, since the peak at 0 prevents uniform convergence.



7. $f_n(x) = x^{1/n}$ on $[0, 1]$ (Discontinuous limit, not uniform)

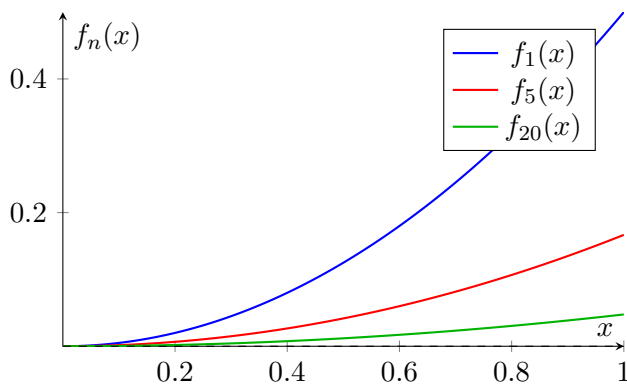
$$f_n(x) \rightarrow f(x) = \begin{cases} 1, & x > 0, \\ 0, & x = 0. \end{cases}$$

Limit function is discontinuous, hence convergence not uniform.



8. $f_n(x) = \frac{x^2}{n+1}$ on $[0, 1]$ (Uniform)

$f_n(x) \rightarrow 0$ uniformly, since $\|f_n\|_\infty = 1/(n+1) \rightarrow 0$.



Uniform Convergence vs. Uniform Continuity

Definitions

Uniform convergence: A sequence of functions $f_n : D \rightarrow \mathbb{R}$ converges *uniformly* to $f : D \rightarrow \mathbb{R}$ if

$$\sup_{x \in D} |f_n(x) - f(x)| \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Uniform continuity: A function $f : D \rightarrow \mathbb{R}$ is *uniformly continuous* on D if

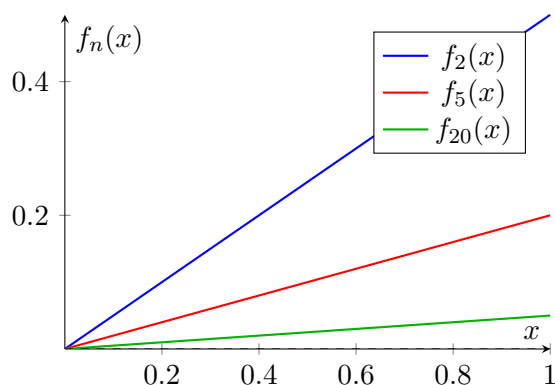
$$\forall \varepsilon > 0 \quad \exists \delta > 0 \quad \text{s.t.} \quad |x - y| < \delta \implies |f(x) - f(y)| < \varepsilon \quad \forall x, y \in D.$$

Key difference

- Uniform convergence concerns *a sequence of functions* approaching a limit.
 - Uniform continuity concerns *a single function* behaving consistently across its domain.
-

Examples of Uniform Convergence

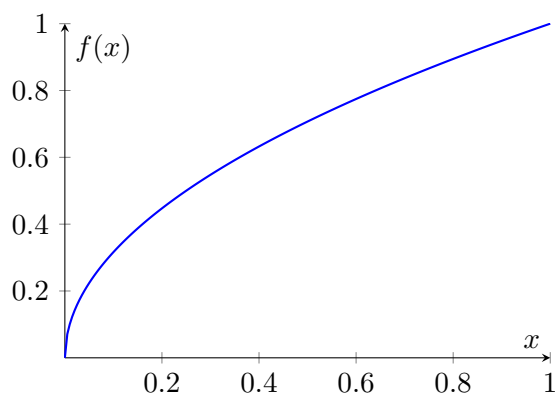
Example 1. $f_n(x) = x/n$ on $[0, 1]$. Then $f_n \rightarrow 0$ uniformly since $\sup_{x \in [0, 1]} |f_n(x)| = 1/n \rightarrow 0$.



Counterexample. $f_n(x) = x^n$ on $[0, 1]$. Pointwise $f(x) = 0$ for $x \in [0, 1)$ and $f(1) = 1$, but not uniform since $\|f_n - f\|_\infty = 1$.

Examples of Uniform Continuity

Example 2. $f(x) = \sqrt{x}$ on $[0, 1]$. Continuous on compact set $\Rightarrow$ uniformly continuous (Heine–Cantor theorem).



Counterexample. $f(x) = x^2$ on $\mathbb{R}$. Continuous but not uniformly continuous: derivative grows without bound, so no global δ works. For example, take $x = n$, $y = n + 1/n$, then $|x - y| = 1/n \rightarrow 0$ but $|x^2 - y^2| \approx 2$.

Summary Table

Concept	Applies to	Key idea
Uniform convergence	Sequence (f_n)	$f_n \rightarrow f$ uniformly if $\sup f_n - f \rightarrow 0$
Uniform continuity	Single function f	One δ works for all $x, y \in D$

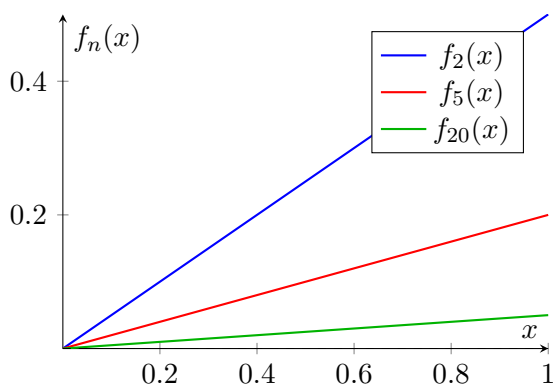
Theorem 1.6 (Cauchy Criterion for Uniform Convergence)

A sequence $(f_i)_{i=0}^{\infty}$ with $f_i \in C^0([a, b])$ converges uniformly to some $f \in C^0([a, b])$ if and only if it is Cauchy in the sup norm:

$$\forall \varepsilon > 0 \quad \exists N \quad \forall m, n \geq N : \sup_{x \in [a, b]} |f_n(x) - f_m(x)| < \varepsilon.$$

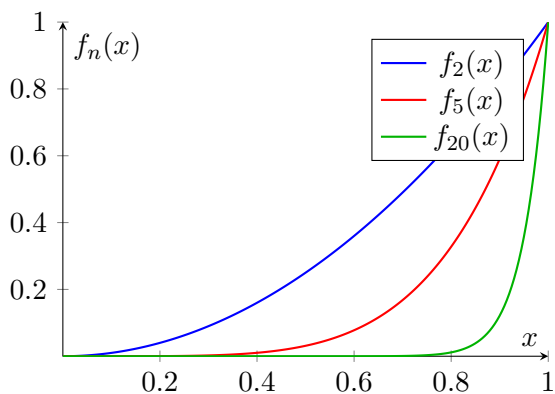
Example 1. Uniform convergence (Cauchy)

$f_n(x) = x/n$ on $[0, 1]$. Limit $f(x) = 0$. Uniform convergence since $\|f_n - 0\|_{\infty} = 1/n \rightarrow 0$.



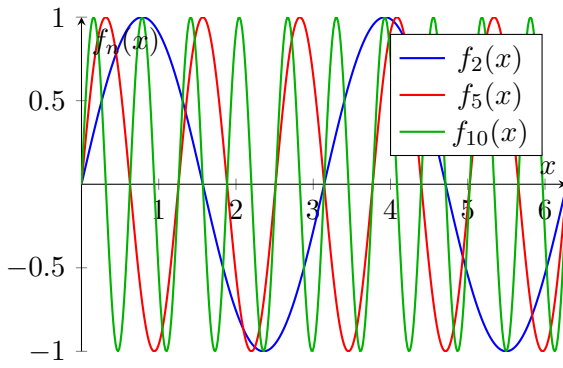
Example 2. Pointwise but not uniform (Not Cauchy)

$f_n(x) = x^n$ on $[0, 1]$. Pointwise limit discontinuous, not uniform, not Cauchy in sup norm.



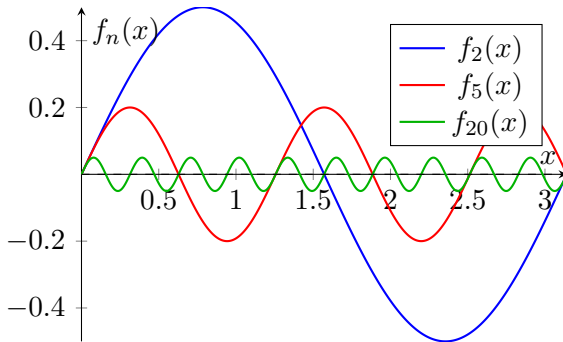
Example 3. Oscillations without convergence (Not Cauchy)

$f_n(x) = \sin(nx)$ on $[0, 2\pi]$. No pointwise limit, not Cauchy.



Example 4. Uniform convergence with oscillations (Cauchy)

$f_n(x) = \sin(nx)/n$ on $[0, \pi]$. Uniform convergence to $f(x) = 0$ since $\|f_n - 0\|_\infty = 1/n \rightarrow 0$.



Theorem 1.7 (Weierstrass M-Test)

Suppose (f_i) is a sequence of functions $f_i : A \rightarrow \mathbb{R}$. If there exists $M_i \geq 0$ such that

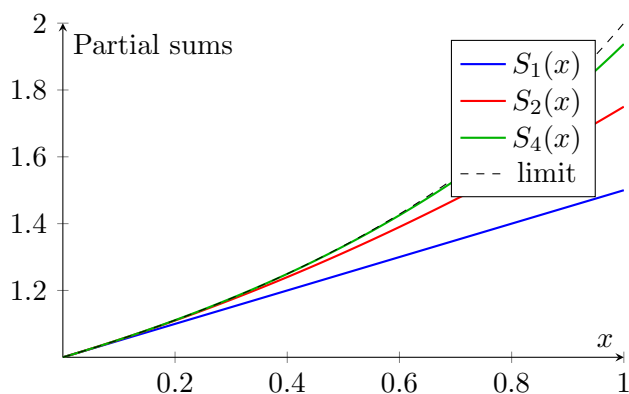
$$|f_i(x)| \leq M_i \quad \forall x \in A, \quad \sum M_i < \infty,$$

then $\sum f_i(x)$ converges uniformly on A .

Example 1: Geometric series of functions

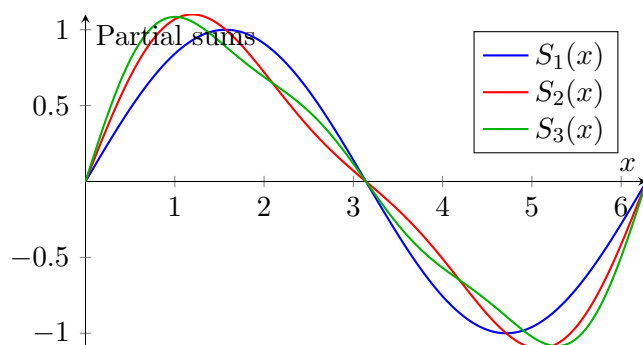
$f_i(x) = \frac{x^i}{2^i}$ on $[0, 1]$. Since $|f_i(x)| \leq 1/2^i$ and $\sum 1/2^i < \infty$, the M-test applies.

$$\sum_{i=0}^{\infty} f_i(x) = \frac{1}{1 - \frac{x}{2}}, \quad x \in [0, 1].$$



Example 2: Trigonometric series

$f_i(x) = \frac{\sin(ix)}{i^2}$, $x \in [0, 2\pi]$. Bounded by $1/i^2$, series $\sum 1/i^2$ converges $\Rightarrow$ uniform convergence.



Counterexample: Divergent bound

$f_i(x) = \frac{\sin(ix)}{i}$ on $[0, 2\pi]$. Here $|f_i(x)| \leq 1/i$, but $\sum 1/i = \infty$. M-test does not apply, and convergence is not uniform.

The Takagi Function

We define

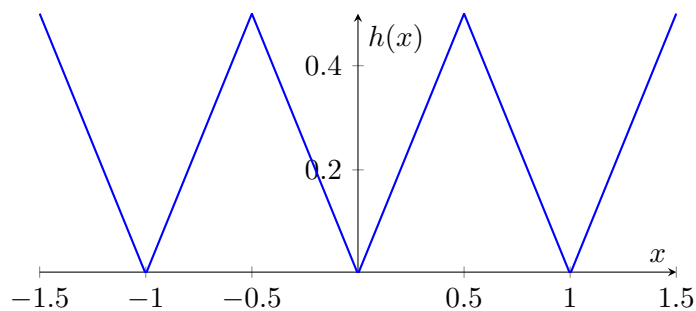
$$h(x) = |x|, \quad x \in \left[-\frac{1}{2}, \frac{1}{2}\right], \quad h(x+1) = h(x),$$

and the Takagi function

$$T(x) = \sum_{i=0}^{\infty} \frac{1}{2^i} h(2^i x).$$

a) The function $h(x)$

$h(x)$ is the periodic “tent map”.



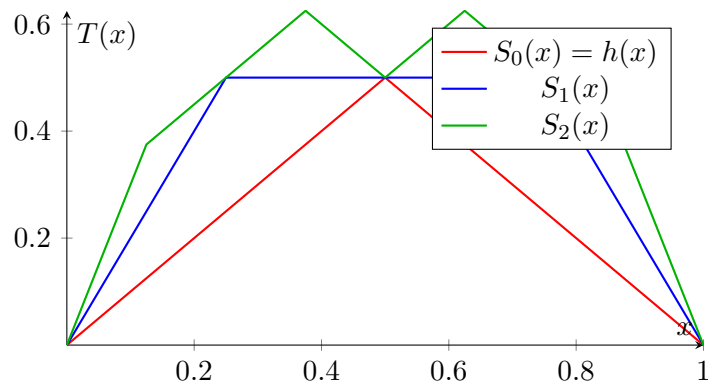
b) Properties of h

- h is uniformly continuous (Lipschitz).
- Differentiable except where $2x \in \mathbb{Z}$ (the corners).

c) The Takagi function $T(x)$

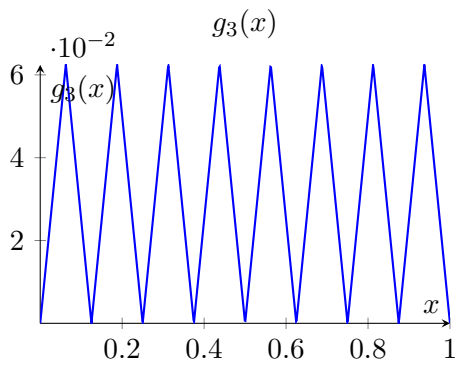
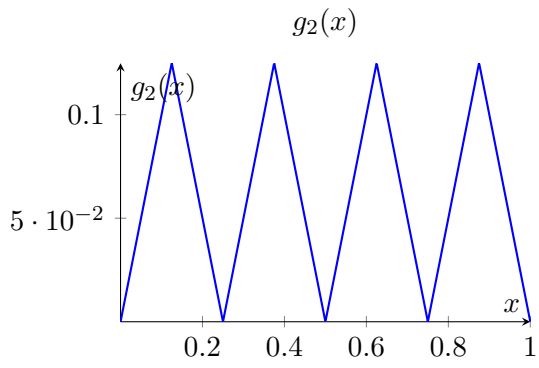
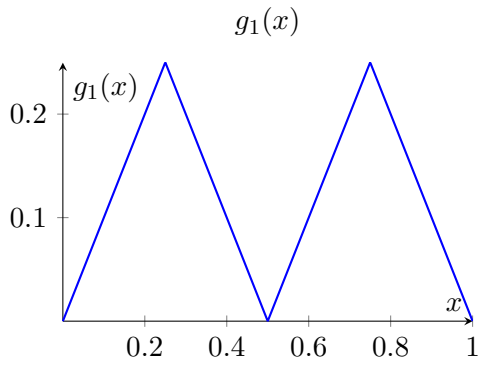
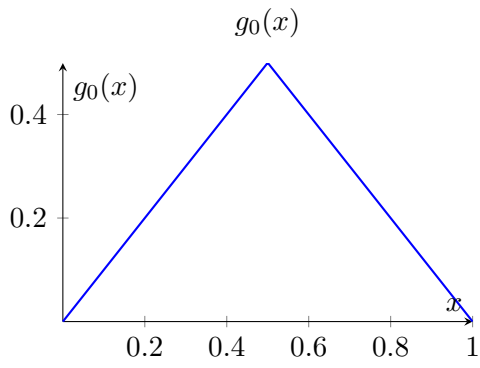
Each term is a rescaled tent function. By the Weierstrass M -test, the series converges uniformly, hence T is continuous.

d) Visualization of partial sums



Components of the Takagi function

We plot $g_i(x) = \frac{1}{2^i} h(2^i x)$ for $i = 0, \dots, 10$ on $[0, 1]$.



Takagi function partial sums $S_n(x)$

We tabulate $S_n(x) = \sum_{i=0}^n \frac{1}{2^i} h(2^i x)$ at dyadic rationals $x = k/2^m$ (where slope changes).

—

Table for $S_0(x) = h(x)$

$$S_0(x) = h(x) = \begin{cases} x, & 0 \leq x \leq 0.5, \\ 1 - x, & 0.5 < x \leq 1. \end{cases}$$

x	0	0.25	0.5	0.75	1
$S_0(x)$	0	0.25	0.5	0.25	0

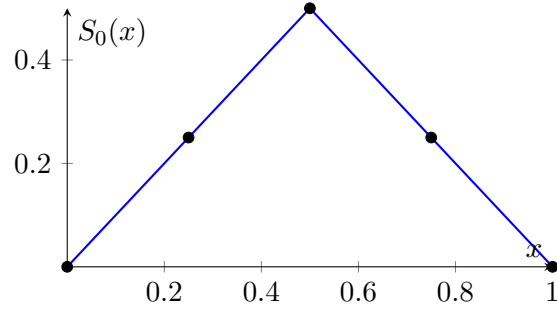


Table for $S_1(x) = h(x) + \frac{1}{2}h(2x)$

$$S_1(x) = h(x) + \frac{1}{2}h(2x).$$

x	0	0.25	0.5	0.75	1
$S_1(x)$	0	0.5	0.5	0.5	0

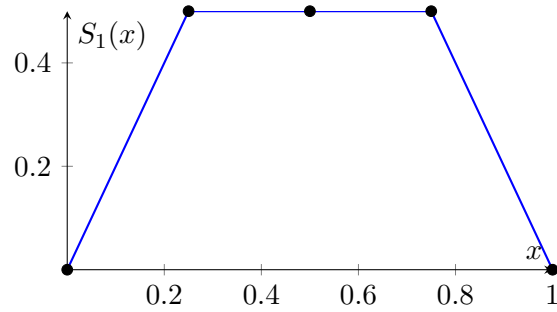


Table for $S_2(x) = h(x) + \frac{1}{2}h(2x) + \frac{1}{4}h(4x)$

$$S_2(x) = h(x) + \frac{1}{2}h(2x) + \frac{1}{4}h(4x).$$

x	0	0.125	0.25	0.375	0.5	0.625	0.75	0.875	1
$S_2(x)$	0	0.375	0.5	0.375	0.5	0.375	0.5	0.375	0

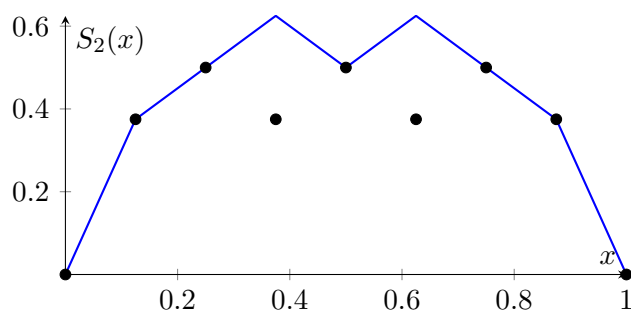
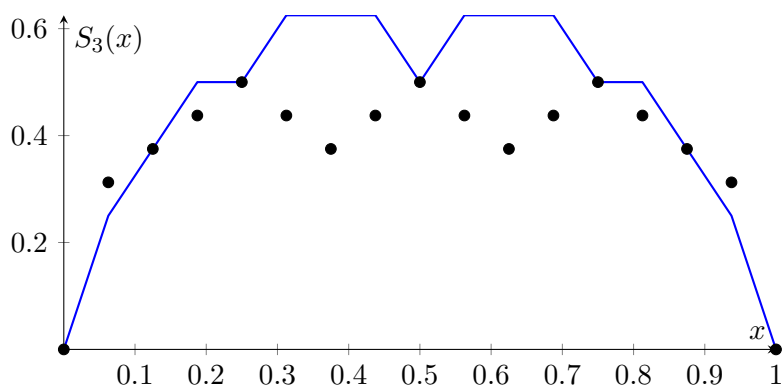


Table for $S_3(x) = S_2(x) + \frac{1}{8}h(8x)$

$$S_3(x) = h(x) + \frac{1}{2}h(2x) + \frac{1}{4}h(4x) + \frac{1}{8}h(8x).$$

x	0	0.0625	0.125	0.1875	0.25	0.3125	0.375	0.4375	0.5	0.5625	0.625	0.6875	0.75	0.8125	0.875	0.9375	1
$S_3(x)$	0	0.3125	0.375	0.4375	0.5	0.4375	0.375	0.4375	0.5	0.4375	0.375	0.4375	0.5	0.4375	0.375	0.4375	0



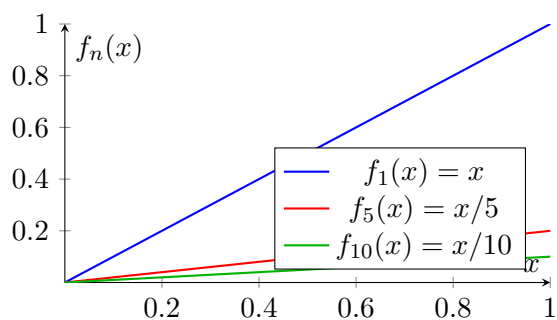
Uniform Equicontinuity

Definition. A family $F = (f_i)$ of functions $f_i : [a, b] \rightarrow \mathbb{R}$ is *uniformly equicontinuous* if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

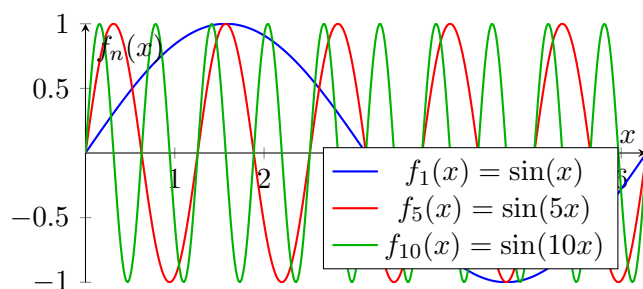
$$|x - y| < \delta \implies |f_i(x) - f_i(y)| < \varepsilon,$$

for all i and all $x, y \in [a, b]$.

Example 1. $f_n(x) = x/n$ on $[0, 1]$ (**Equicontinuous**)

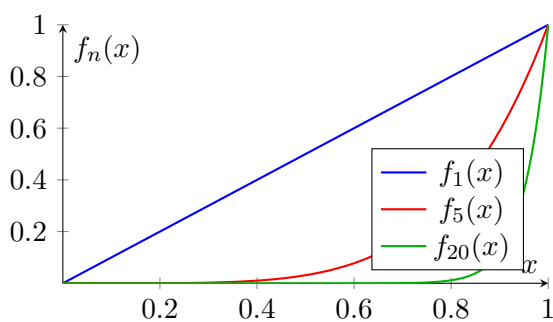


Example 2. $f_n(x) = \sin(nx)$ on $[0, 2\pi]$ (Not Equicontinuous)

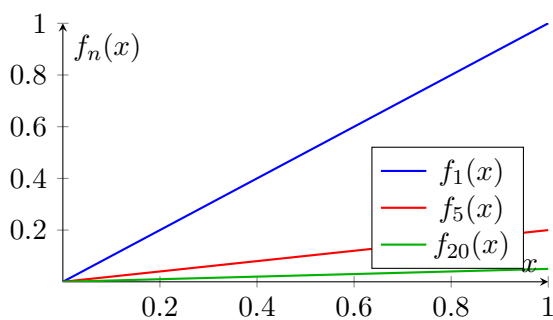


Examples for Arzelà–Ascoli Theorem

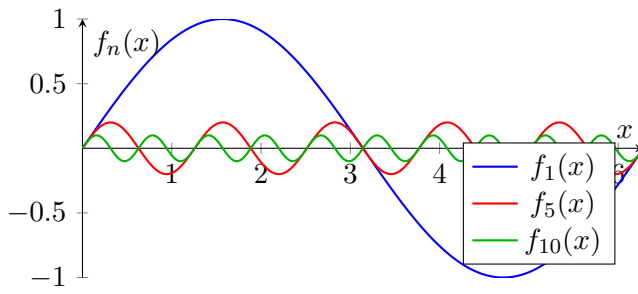
Example 1: $f_n(x) = x^n$ on $[0, 1]$ (Not equicontinuous)



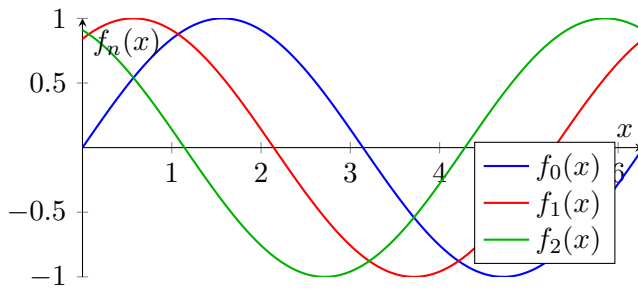
Example 2: $f_n(x) = x/n$ on $[0, 1]$ (Equicontinuous, converges uniformly to 0)



Example 3: $f_n(x) = \sin(nx)/n$ on $[0, 2\pi]$ (Equicontinuous, converges uniformly to 0)



Example 4: $f_n(x) = \sin(x + n)$ on $[0, 2\pi]$ (Subsequences converge uniformly)



Exercise 1.1

Consider the meromorphic function

$$F(z) = \frac{1}{1+z^2} \frac{\cos[(\pi-\theta)z]}{2\sin(\pi z)}.$$

a) Poles and residues

The poles occur at $z = \pm i$ and $z = n$, $n \in \mathbb{Z}$. At $z = \pm i$:

$$\text{Res}[F, \pm i] = -\frac{\cosh(\pi - \theta)}{4 \sinh \pi}.$$

At $z = n \in \mathbb{Z}$:

$$\text{Res}[F, n] = \frac{\cos(n\theta)}{2\pi(1+n^2)}.$$

b) The contour $\Gamma^{(N)}$

Define

$$\Gamma_1^{(N)}(t) = \left(N + \frac{1}{2}\right)(1 + it), \quad -1 \leq t \leq 1,$$

$$\Gamma_2^{(N)}(t) = \left(N + \frac{1}{2}\right)(-t + i), \quad -1 \leq t \leq 1,$$

$$\Gamma_3^{(N)}(t) = \left(N + \frac{1}{2}\right)(-1 - it), \quad -1 \leq t \leq 1,$$

$$\Gamma_4^{(N)}(t) = \left(N + \frac{1}{2}\right)(t - i), \quad -1 \leq t \leq 1.$$

Then $\Gamma^{(N)}$ is the closed square contour with vertices $\pm(N + \frac{1}{2}) \pm i(N + \frac{1}{2})$. It encloses the poles at $\pm i$ and all integers n with $|n| \leq N$.

c) Bound on the contour integral

On $\Gamma^{(N)}$, $|z| \sim N$, so $|F(z)| = O(1/N^2)$. Since the contour has length $O(N)$, we obtain

$$\left| \int_{\Gamma^{(N)}} F(z) dz \right| \leq \frac{C}{N},$$

for some constant C independent of N .

d) Residue theorem and Fourier series

By Cauchy's residue theorem,

$$\int_{\Gamma^{(N)}} F(z) dz = 2\pi i \left(\text{Res}[F, i] + \text{Res}[F, -i] + \sum_{n=-N}^N \text{Res}[F, n] \right).$$

Hence

$$\left| \frac{\cosh(\pi - \theta)}{2 \sinh \pi} - \sum_{n=-N}^N \frac{\cos(n\theta)}{1 + n^2} \right| \leq \frac{C}{N}.$$

Letting $N \rightarrow \infty$, we conclude

$$\frac{\cosh(\pi - \theta)}{2 \sinh \pi} = \sum_{n=-\infty}^{\infty} \frac{e^{in\theta}}{1 + n^2}.$$

This gives the Fourier series expansion, uniformly in θ .